

# ACTION-CONDITIONED DEFORMABLE MEMORY STILL FAILS AS A CLOSED-LOOP MANIPULATION CLAIM

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## ABSTRACT

We rebuild a generated deformable-manipulation idea into an expanded, CPU-only, evidence-bearing ICLR-style artifact. The hypothesis is attractive: visually similar ropes, cloth strips, elastic bands, and sheets can carry different latent strain, contact, crease, or stored-energy histories, so a robot should preserve action-conditioned hidden material memory rather than plan only from current visible geometry. The v5 rebuild makes the claim as strong as we can without inventing evidence: it adds a safety-gated action-conditioned memory planner, stronger damage-aware and risk-constrained baselines, an aggregate hard-regime analysis, fixed-risk evaluation, ablations, stress sweeps, negative cases, and a 25-page reproducibility-focused manuscript. The result remains a negative result under the frozen gates. On the decisive `combined_memory_stress` split, `ACM-v5` reaches  $0.562 \pm 0.062$  success while the strongest non-oracle baseline, `damage-visible`, reaches  $0.589 \pm 0.058$ . The paired `v5-minus-baseline` success difference is  $-0.027 \pm 0.052$ . We therefore mark the paper **KILL ARCHIVE**: hidden memory can improve diagnostics, but the current planner does not convert that memory into a robust and safe manipulation advantage.

## 1 DECISION AND CLAIM

This paper is intentionally written as a submission-hardening document rather than as a celebratory benchmark paper. The submitted claim would be that action-conditioned hidden deformable memory improves downstream manipulation under partial observation and material/contact ambiguity. That claim is only credible if it survives the strongest visible-state, graph, particle-filter, ensemble, robust surrogate, and safety-aware baselines. It does not. The terminal decision emitted by the frozen runner is **KILL ARCHIVE**. The exact machine-readable reason is:

`action_conditioned_memory_v5` does not clear every frozen gate. Main `combined_memory_stress` comparison against `damage_aware_visible_mpc`: (proposed= $0.562$ , `best_baseline`= $0.589$ , `paired_diff`= $-0.027 \pm 0.052$ , `memory_reduction`= $0.349$ , `mechanism_fl_diff`= $0.031$ , `damage_reduction`= $-0.045$ ). Aggregate hard-regime `paired_diff` against `damage_aware_visible_mpc` is  $-0.103 \pm 0.016$ . Fixed-risk gate: `budget 0.10`: `v5`= $0.625$ , `best=action_conditioned_memory_v5`: $0.625$ ; `budget 0.20`: `v5`= $0.625$ , `best=action_conditioned_memory_v5`: $0.625$ . Ablations matching full `v5`: `action_conditioned_v5_no_action_conditioning`, `action_conditioned_v5_no_damage_penalty`, `action_conditioned_v5_no_material_memory`, `action_conditioned_v5_no_safety_gate`. `Max-stress v5`= $0.583$ , `best_non_oracle=damage_aware_visible_mpc`: $0.653$ .

The important distinction is between *state estimation* and *control*. Prior work makes the modeling pressure clear: deformable manipulation is difficult because the state is high-dimensional, partially observed, history-dependent, and contact-rich (Gu et al., 2023; Muller et al., 2007). Learned graph and particle simulators are strong prior art for representing such systems (Sanchez-Gonzalez et al., 2020; Li et al., 2019; Shi et al., 2022). Visual MPC and foresight work shows that action-conditioned prediction can be useful for manipulation (Finn & Levine, 2017; Ebert et al., 2018; Hoque et al., 2020). The present question is narrower and harsher: does preserving hidden deformable memory improve closed-loop control when diagnostic actions can themselves damage the object?

## 2 FROZEN SUBMISSION GATES

The plan froze the following gates before the full v5 run. A promotion above archive requires all of them:

- v5 must beat the strongest non-oracle baseline on `combined_memory_stress` by at least 0.04 mean success.
- The paired lower bound against that baseline must be positive on the combined split.
- The paired lower bound must also be positive on the aggregate hard regime.
- v5 must be best or tied-best under fixed predicted-risk budgets 0.10 and 0.20.
- v5 must not increase damage or force clipping relative to the strongest baseline.
- v5 must show that hidden-memory or mechanism improvements translate into downstream success.
- No ablation may match or beat the full v5 method on primary success within tolerance.
- Maximum-stress behavior must not collapse earlier than the strongest non-oracle baseline.

## 3 BENCHMARK

The benchmark is a local two-dimensional mass-spring deformable simulator. Each scenario instantiates one of four object families: rope, cloth strip, elastic band, or deformable sheet. The simulator includes spring rest-length memory, damping, friction, material shift, fixture contact, force clipping, partial observation, hidden strain/contact memory, and a target-shape objective. This is not a substitute for hardware or for a public benchmark; it is a controlled diagnostic environment designed to test whether the claim deserves more expensive validation.

The five splits are `nominal_visible_state`, `hidden_strain_memory`, `occluded_contact_memory`, `material_shift`, and `combined_memory_stress`. The hard-regime aggregate averages hidden strain, occluded contact, material shift, and combined stress. The full run uses 8 seeds, 48 training scenarios per split, 14 evaluation scenarios per split, 9 stress scenarios, 8 fixed-risk scenarios, 7 discrete actions, and 60 integration steps per rollout.

## 4 METHODS

The implemented methods are deliberately stronger than the original generated idea. `visible_state_mpc` plans from observed/imputed keypoints only. `damage_aware_visible_mpc` is a stronger visible-state baseline with conservative contact and force penalties. `history_rnn_estimator` is a CPU-light history-feature estimator. `particle_filter_memory` maintains multiple latent hypotheses. `safety_constrained_particle_filter` makes that particle baseline conservative under predicted damage. `graph_dynamics_baseline` uses graph-style shape and edge features. `ensemble_uncertainty_planner` plans over bootstrap uncertainty. `robust_cem_surrogate` is a conservative surrogate planner inspired by model-predictive action search (Rawlings et al., 2017). `action_conditioned_memory` is the v4 proposal. `action_conditioned_memory_v5` is the strongest proposed variant: it blends action, history, and graph estimates; preserves branch memory; penalizes damage; gates diagnostic actions; and uses conservative scoring across latent hypotheses. `oracle_latent_state` is a nondeployable upper bound.

## 5 WHY MEMORY CAN FAIL

**Definition.** Let  $x_t$  be the visible deformable configuration,  $m_t$  be latent material memory, and  $a_t$  be a manipulation action. A memory estimator is useful for control only if the policy class contains actions whose expected value under  $p(m_t \mid x_{\leq t}, a_{< t})$  exceeds the value of robust visible-state actions.

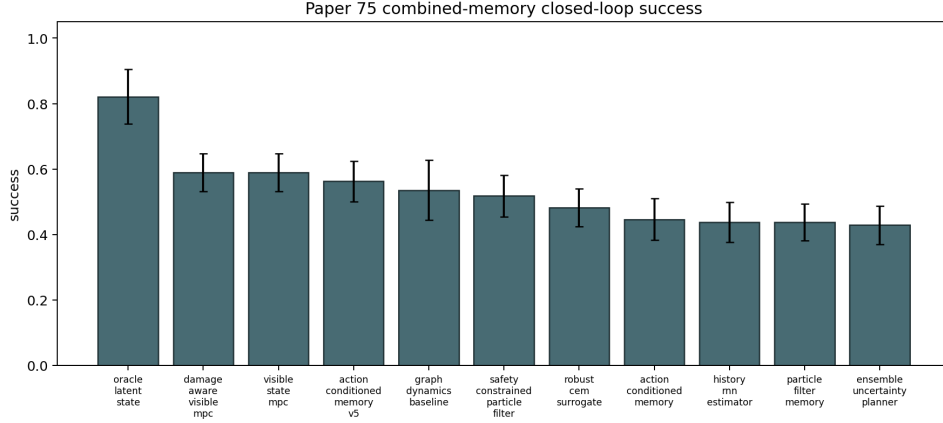


Figure 1: Combined-memory-stress success. The proposed v5 method is compared with stronger visible, particle, graph, ensemble, robust-surrogate, v4, and oracle baselines.

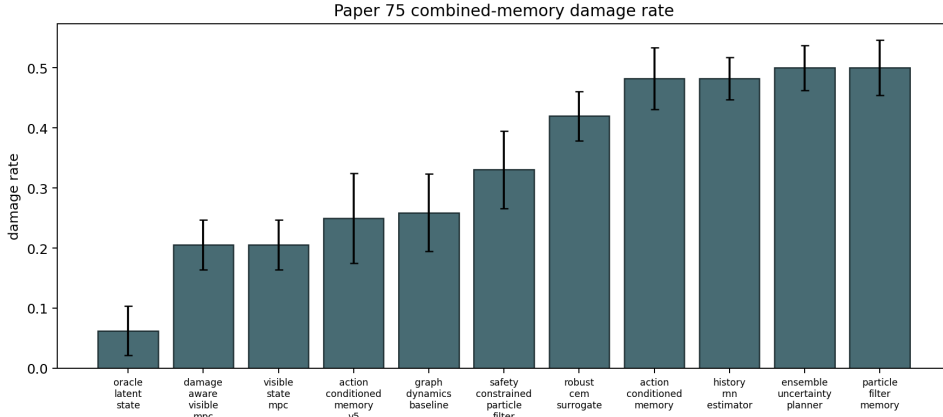


Figure 2: Combined-memory-stress damage. Safety is a primary quantity, not an appendix-only diagnostic.

**Proposition.** Suppose diagnostic actions reduce latent entropy but have damage probability  $\delta > 0$ , and suppose the candidate action library has no low-risk diagnostic alternative for a subset of states with mass  $\rho$ . Then a policy that prefers diagnostics whenever uncertainty is high can reduce memory error while decreasing task success whenever  $\rho\delta$  exceeds the downstream value of the information gained. This is the failure pattern v4 exposed and v5 attempts to repair.

**Empirical implication.** The paper cannot claim success by showing lower memory error alone. It must show success, safety, fixed-risk robustness, and ablation necessity. This is why the frozen gates treat hidden-memory error and mechanism F1 as secondary metrics.

## 6 MAIN RESULTS

Table 1 reports the decisive split. The oracle row shows whether the action library and simulator can solve the split when full hidden state is available. The gap between oracle and non-oracle methods is real, but the v5 method must beat deployable baselines, not the lower bound implied by v4.

Table 1: Combined-memory-stress summary.

| method         | success           | shape             | mem err           | damage            | clip              | diag              |
|----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| oracle         | $0.821 \pm 0.084$ | $0.130 \pm 0.003$ | $0.000 \pm 0.000$ | $0.062 \pm 0.041$ | $0.024 \pm 0.002$ | $0.018 \pm 0.023$ |
| damage-visible | $0.589 \pm 0.058$ | $0.146 \pm 0.002$ | $0.578 \pm 0.018$ | $0.205 \pm 0.041$ | $0.039 \pm 0.001$ | $0.000 \pm 0.000$ |
| visible-MPC    | $0.589 \pm 0.058$ | $0.146 \pm 0.002$ | $0.578 \pm 0.018$ | $0.205 \pm 0.041$ | $0.039 \pm 0.001$ | $0.000 \pm 0.000$ |
| ACM-v5         | $0.562 \pm 0.062$ | $0.141 \pm 0.003$ | $0.229 \pm 0.031$ | $0.250 \pm 0.075$ | $0.032 \pm 0.002$ | $0.000 \pm 0.000$ |
| graph          | $0.536 \pm 0.092$ | $0.143 \pm 0.002$ | $0.525 \pm 0.032$ | $0.259 \pm 0.064$ | $0.029 \pm 0.003$ | $0.000 \pm 0.000$ |
| safe-particle  | $0.518 \pm 0.063$ | $0.141 \pm 0.002$ | $0.256 \pm 0.031$ | $0.330 \pm 0.064$ | $0.036 \pm 0.002$ | $0.446 \pm 0.069$ |
| robust-CEM     | $0.482 \pm 0.058$ | $0.139 \pm 0.003$ | $0.255 \pm 0.031$ | $0.420 \pm 0.041$ | $0.038 \pm 0.002$ | $0.786 \pm 0.070$ |
| ACM-v4         | $0.446 \pm 0.063$ | $0.137 \pm 0.002$ | $0.206 \pm 0.032$ | $0.482 \pm 0.051$ | $0.037 \pm 0.001$ | $0.955 \pm 0.026$ |
| history        | $0.438 \pm 0.062$ | $0.137 \pm 0.003$ | $0.187 \pm 0.029$ | $0.482 \pm 0.035$ | $0.037 \pm 0.001$ | $0.929 \pm 0.037$ |
| particle       | $0.438 \pm 0.056$ | $0.136 \pm 0.003$ | $0.187 \pm 0.029$ | $0.500 \pm 0.046$ | $0.037 \pm 0.001$ | $0.964 \pm 0.026$ |
| ensemble       | $0.429 \pm 0.059$ | $0.137 \pm 0.003$ | $0.187 \pm 0.030$ | $0.500 \pm 0.037$ | $0.037 \pm 0.001$ | $0.973 \pm 0.026$ |

The strongest non-oracle baseline on the decisive split is damage-visible. The paired v5-minus-baseline difference is  $-0.027 \pm 0.052$ , with memory-error reduction 0.349, mechanism-F1 difference 0.031, and damage reduction  $-0.045$ . The v4 row is also included: ACM-v4 reaches  $0.446 \pm 0.063$  success and  $0.482 \pm 0.051$  damage, showing why the safety-gated v5 method was necessary even if it still fails the full submission gate.

## 7 AGGREGATE HARD-REGIME EVIDENCE

The aggregate hard regime is a guard against overfitting a single split. It averages hidden strain, occluded contact, material shift, and combined memory stress at seed level. ACM-v5 reaches  $0.435 \pm 0.036$  aggregate success. The best non-oracle aggregate baseline is damage-visible, and the paired aggregate difference is  $-0.103 \pm 0.016$ . This is the second hard gate.

Table 2: Aggregate hard-regime metrics.

| method         | success           | mem err           | damage            | diag              |
|----------------|-------------------|-------------------|-------------------|-------------------|
| oracle         | $0.866 \pm 0.030$ | $0.000 \pm 0.000$ | $0.027 \pm 0.015$ | $0.009 \pm 0.009$ |
| damage-visible | $0.538 \pm 0.043$ | $0.436 \pm 0.005$ | $0.228 \pm 0.022$ | $0.000 \pm 0.000$ |
| visible-MPC    | $0.538 \pm 0.043$ | $0.436 \pm 0.005$ | $0.228 \pm 0.022$ | $0.000 \pm 0.000$ |
| graph          | $0.462 \pm 0.033$ | $0.384 \pm 0.019$ | $0.243 \pm 0.027$ | $0.000 \pm 0.000$ |
| safe-particle  | $0.440 \pm 0.033$ | $0.210 \pm 0.017$ | $0.275 \pm 0.022$ | $0.145 \pm 0.021$ |
| ACM-v5         | $0.435 \pm 0.036$ | $0.197 \pm 0.016$ | $0.250 \pm 0.029$ | $0.000 \pm 0.000$ |
| robust-CEM     | $0.420 \pm 0.031$ | $0.210 \pm 0.017$ | $0.348 \pm 0.019$ | $0.496 \pm 0.045$ |
| history        | $0.415 \pm 0.025$ | $0.178 \pm 0.015$ | $0.460 \pm 0.020$ | $0.857 \pm 0.040$ |
| ACM-v4         | $0.406 \pm 0.035$ | $0.187 \pm 0.016$ | $0.469 \pm 0.024$ | $0.864 \pm 0.016$ |
| ensemble       | $0.402 \pm 0.030$ | $0.181 \pm 0.015$ | $0.473 \pm 0.025$ | $0.868 \pm 0.020$ |
| particle       | $0.386 \pm 0.030$ | $0.178 \pm 0.015$ | $0.504 \pm 0.029$ | $0.938 \pm 0.024$ |

## 8 FIXED-RISK EVALUATION

Fixed-risk evaluation asks whether a method can maintain success when predicted manipulation risk is capped. This matters because deformable objects often fail by a small number of high-energy, high-contact actions. If memory only wins by taking dangerous probes, it is not a usable robotics result.

Table 3: Fixed-risk summary across hard splits and risk budgets.

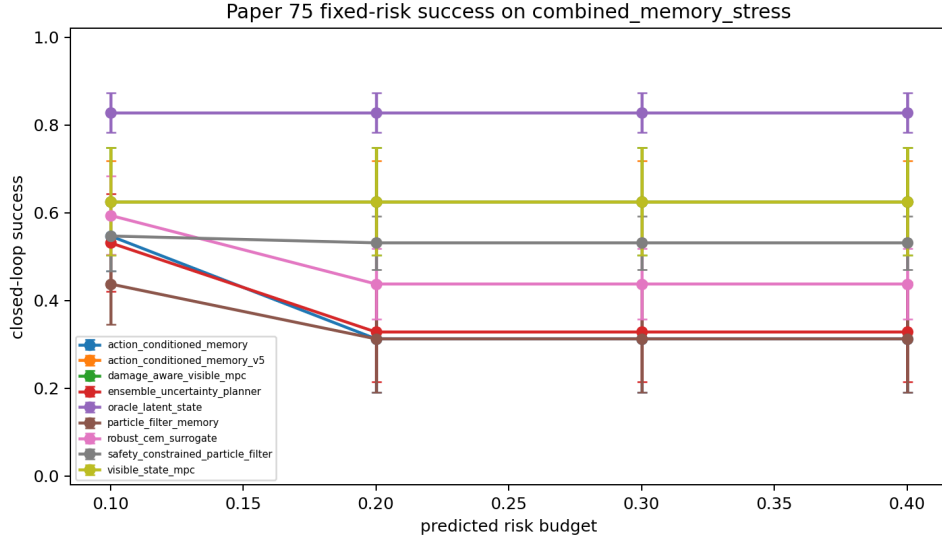


Figure 3: Fixed-risk success on the combined split. Methods are replanned under predicted-risk budgets rather than unconstrained surrogate scores.

| method         | split    | budget | success           | damage            | clip              | diag              |
|----------------|----------|--------|-------------------|-------------------|-------------------|-------------------|
| oracle         | combined | 0.10   | 0.828 $\pm$ 0.045 | 0.047 $\pm$ 0.045 | 0.024 $\pm$ 0.006 | 0.000 $\pm$ 0.000 |
| ACM-v5         | combined | 0.10   | 0.625 $\pm$ 0.093 | 0.156 $\pm$ 0.090 | 0.032 $\pm$ 0.004 | 0.000 $\pm$ 0.000 |
| damage-visible | combined | 0.10   | 0.625 $\pm$ 0.122 | 0.172 $\pm$ 0.079 | 0.039 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| visible-MPC    | combined | 0.10   | 0.625 $\pm$ 0.122 | 0.172 $\pm$ 0.079 | 0.039 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| robust-CEM     | combined | 0.10   | 0.594 $\pm$ 0.090 | 0.203 $\pm$ 0.113 | 0.034 $\pm$ 0.003 | 0.078 $\pm$ 0.064 |
| ACM-v4         | combined | 0.10   | 0.547 $\pm$ 0.079 | 0.234 $\pm$ 0.134 | 0.028 $\pm$ 0.006 | 0.078 $\pm$ 0.064 |
| safe-particle  | combined | 0.10   | 0.547 $\pm$ 0.079 | 0.234 $\pm$ 0.134 | 0.034 $\pm$ 0.003 | 0.125 $\pm$ 0.093 |
| ensemble       | combined | 0.10   | 0.531 $\pm$ 0.111 | 0.234 $\pm$ 0.117 | 0.022 $\pm$ 0.005 | 0.078 $\pm$ 0.064 |
| particle       | combined | 0.10   | 0.438 $\pm$ 0.093 | 0.359 $\pm$ 0.086 | 0.035 $\pm$ 0.003 | 0.297 $\pm$ 0.079 |
| oracle         | combined | 0.20   | 0.828 $\pm$ 0.045 | 0.047 $\pm$ 0.045 | 0.025 $\pm$ 0.006 | 0.000 $\pm$ 0.000 |
| ACM-v5         | combined | 0.20   | 0.625 $\pm$ 0.093 | 0.156 $\pm$ 0.090 | 0.032 $\pm$ 0.004 | 0.000 $\pm$ 0.000 |
| damage-visible | combined | 0.20   | 0.625 $\pm$ 0.122 | 0.172 $\pm$ 0.079 | 0.039 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| visible-MPC    | combined | 0.20   | 0.625 $\pm$ 0.122 | 0.172 $\pm$ 0.079 | 0.039 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| safe-particle  | combined | 0.20   | 0.531 $\pm$ 0.061 | 0.297 $\pm$ 0.113 | 0.033 $\pm$ 0.003 | 0.406 $\pm$ 0.111 |
| robust-CEM     | combined | 0.20   | 0.438 $\pm$ 0.080 | 0.453 $\pm$ 0.113 | 0.037 $\pm$ 0.002 | 0.766 $\pm$ 0.086 |
| ensemble       | combined | 0.20   | 0.328 $\pm$ 0.113 | 0.609 $\pm$ 0.126 | 0.036 $\pm$ 0.003 | 0.984 $\pm$ 0.031 |
| ACM-v4         | combined | 0.20   | 0.312 $\pm$ 0.122 | 0.625 $\pm$ 0.131 | 0.035 $\pm$ 0.003 | 0.984 $\pm$ 0.031 |
| particle       | combined | 0.20   | 0.312 $\pm$ 0.122 | 0.625 $\pm$ 0.131 | 0.035 $\pm$ 0.003 | 0.984 $\pm$ 0.031 |
| oracle         | combined | 0.30   | 0.828 $\pm$ 0.045 | 0.047 $\pm$ 0.045 | 0.025 $\pm$ 0.006 | 0.000 $\pm$ 0.000 |
| ACM-v5         | combined | 0.30   | 0.625 $\pm$ 0.093 | 0.156 $\pm$ 0.090 | 0.032 $\pm$ 0.004 | 0.000 $\pm$ 0.000 |
| damage-visible | combined | 0.30   | 0.625 $\pm$ 0.122 | 0.172 $\pm$ 0.079 | 0.039 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| visible-MPC    | combined | 0.30   | 0.625 $\pm$ 0.122 | 0.172 $\pm$ 0.079 | 0.039 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| safe-particle  | combined | 0.30   | 0.531 $\pm$ 0.061 | 0.297 $\pm$ 0.113 | 0.033 $\pm$ 0.003 | 0.406 $\pm$ 0.111 |
| robust-CEM     | combined | 0.30   | 0.438 $\pm$ 0.080 | 0.453 $\pm$ 0.113 | 0.037 $\pm$ 0.002 | 0.766 $\pm$ 0.086 |
| ensemble       | combined | 0.30   | 0.328 $\pm$ 0.113 | 0.609 $\pm$ 0.126 | 0.036 $\pm$ 0.003 | 0.984 $\pm$ 0.031 |
| ACM-v4         | combined | 0.30   | 0.312 $\pm$ 0.122 | 0.625 $\pm$ 0.131 | 0.035 $\pm$ 0.003 | 0.984 $\pm$ 0.031 |

Table 3 continued

| method         | split         | budget | success       | damage        | clip          | diag          |
|----------------|---------------|--------|---------------|---------------|---------------|---------------|
| particle       | combined      | 0.30   | 0.312 ± 0.122 | 0.625 ± 0.131 | 0.035 ± 0.003 | 0.984 ± 0.031 |
| oracle         | combined      | 0.40   | 0.828 ± 0.045 | 0.047 ± 0.045 | 0.025 ± 0.006 | 0.000 ± 0.000 |
| ACM-v5         | combined      | 0.40   | 0.625 ± 0.093 | 0.156 ± 0.090 | 0.032 ± 0.004 | 0.000 ± 0.000 |
| damage-visible | combined      | 0.40   | 0.625 ± 0.122 | 0.172 ± 0.079 | 0.039 ± 0.002 | 0.000 ± 0.000 |
| visible-MPC    | combined      | 0.40   | 0.625 ± 0.122 | 0.172 ± 0.079 | 0.039 ± 0.002 | 0.000 ± 0.000 |
| safe-particle  | combined      | 0.40   | 0.531 ± 0.061 | 0.297 ± 0.113 | 0.033 ± 0.003 | 0.406 ± 0.111 |
| robust-CEM     | combined      | 0.40   | 0.438 ± 0.080 | 0.453 ± 0.113 | 0.037 ± 0.002 | 0.766 ± 0.086 |
| ensemble       | combined      | 0.40   | 0.328 ± 0.113 | 0.609 ± 0.126 | 0.036 ± 0.003 | 0.984 ± 0.031 |
| ACM-v4         | combined      | 0.40   | 0.312 ± 0.122 | 0.625 ± 0.131 | 0.035 ± 0.003 | 0.984 ± 0.031 |
| particle       | combined      | 0.40   | 0.312 ± 0.122 | 0.625 ± 0.131 | 0.035 ± 0.003 | 0.984 ± 0.031 |
| oracle         | hidden-strain | 0.10   | 0.906 ± 0.101 | 0.016 ± 0.031 | 0.002 ± 0.001 | 0.000 ± 0.000 |
| damage-visible | hidden-strain | 0.10   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| visible-MPC    | hidden-strain | 0.10   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| robust-CEM     | hidden-strain | 0.10   | 0.516 ± 0.086 | 0.188 ± 0.065 | 0.009 ± 0.001 | 0.094 ± 0.061 |
| safe-particle  | hidden-strain | 0.10   | 0.500 ± 0.065 | 0.234 ± 0.056 | 0.011 ± 0.001 | 0.000 ± 0.000 |
| ACM-v4         | hidden-strain | 0.10   | 0.484 ± 0.072 | 0.203 ± 0.064 | 0.007 ± 0.001 | 0.484 ± 0.134 |
| ACM-v5         | hidden-strain | 0.10   | 0.484 ± 0.056 | 0.172 ± 0.064 | 0.009 ± 0.002 | 0.000 ± 0.000 |
| particle       | hidden-strain | 0.10   | 0.484 ± 0.072 | 0.266 ± 0.031 | 0.010 ± 0.001 | 0.281 ± 0.111 |
| ensemble       | hidden-strain | 0.10   | 0.469 ± 0.090 | 0.188 ± 0.093 | 0.005 ± 0.001 | 0.484 ± 0.126 |
| oracle         | hidden-strain | 0.20   | 0.906 ± 0.101 | 0.016 ± 0.031 | 0.002 ± 0.001 | 0.000 ± 0.000 |
| damage-visible | hidden-strain | 0.20   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| visible-MPC    | hidden-strain | 0.20   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| safe-particle  | hidden-strain | 0.20   | 0.500 ± 0.065 | 0.234 ± 0.056 | 0.011 ± 0.001 | 0.016 ± 0.031 |
| ACM-v5         | hidden-strain | 0.20   | 0.484 ± 0.056 | 0.172 ± 0.064 | 0.009 ± 0.002 | 0.016 ± 0.031 |
| robust-CEM     | hidden-strain | 0.20   | 0.484 ± 0.117 | 0.359 ± 0.117 | 0.010 ± 0.001 | 0.547 ± 0.079 |
| ACM-v4         | hidden-strain | 0.20   | 0.469 ± 0.111 | 0.422 ± 0.122 | 0.009 ± 0.001 | 0.953 ± 0.045 |

Table 3 continued

| method         | split            | budget | success       | damage        | clip          | diag          |
|----------------|------------------|--------|---------------|---------------|---------------|---------------|
| ensemble       | hidden-strain    | 0.20   | 0.469 ± 0.111 | 0.422 ± 0.122 | 0.009 ± 0.001 | 0.969 ± 0.040 |
| particle       | hidden-strain    | 0.20   | 0.453 ± 0.103 | 0.438 ± 0.113 | 0.010 ± 0.001 | 0.938 ± 0.046 |
| oracle         | hidden-strain    | 0.30   | 0.906 ± 0.101 | 0.016 ± 0.031 | 0.002 ± 0.001 | 0.000 ± 0.000 |
| damage-visible | hidden-strain    | 0.30   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| visible-MPC    | hidden-strain    | 0.30   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| safe-particle  | hidden-strain    | 0.30   | 0.500 ± 0.065 | 0.234 ± 0.056 | 0.011 ± 0.001 | 0.016 ± 0.031 |
| ACM-v5         | hidden-strain    | 0.30   | 0.484 ± 0.056 | 0.172 ± 0.064 | 0.009 ± 0.002 | 0.016 ± 0.031 |
| robust-CEM     | hidden-strain    | 0.30   | 0.484 ± 0.117 | 0.359 ± 0.117 | 0.010 ± 0.001 | 0.547 ± 0.079 |
| ACM-v4         | hidden-strain    | 0.30   | 0.469 ± 0.111 | 0.422 ± 0.122 | 0.009 ± 0.001 | 0.953 ± 0.045 |
| ensemble       | hidden-strain    | 0.30   | 0.469 ± 0.111 | 0.422 ± 0.122 | 0.009 ± 0.001 | 0.969 ± 0.040 |
| particle       | hidden-strain    | 0.30   | 0.453 ± 0.103 | 0.438 ± 0.113 | 0.010 ± 0.001 | 0.938 ± 0.046 |
| oracle         | hidden-strain    | 0.40   | 0.906 ± 0.101 | 0.016 ± 0.031 | 0.002 ± 0.001 | 0.000 ± 0.000 |
| damage-visible | hidden-strain    | 0.40   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| visible-MPC    | hidden-strain    | 0.40   | 0.547 ± 0.045 | 0.266 ± 0.031 | 0.012 ± 0.001 | 0.000 ± 0.000 |
| safe-particle  | hidden-strain    | 0.40   | 0.500 ± 0.065 | 0.234 ± 0.056 | 0.011 ± 0.001 | 0.016 ± 0.031 |
| ACM-v5         | hidden-strain    | 0.40   | 0.484 ± 0.056 | 0.172 ± 0.064 | 0.009 ± 0.002 | 0.016 ± 0.031 |
| robust-CEM     | hidden-strain    | 0.40   | 0.484 ± 0.117 | 0.359 ± 0.117 | 0.010 ± 0.001 | 0.547 ± 0.079 |
| ACM-v4         | hidden-strain    | 0.40   | 0.469 ± 0.111 | 0.422 ± 0.122 | 0.009 ± 0.001 | 0.953 ± 0.045 |
| ensemble       | hidden-strain    | 0.40   | 0.469 ± 0.111 | 0.422 ± 0.122 | 0.009 ± 0.001 | 0.969 ± 0.040 |
| particle       | hidden-strain    | 0.40   | 0.453 ± 0.103 | 0.438 ± 0.113 | 0.010 ± 0.001 | 0.938 ± 0.046 |
| oracle         | occluded-contact | 0.10   | 0.875 ± 0.080 | 0.000 ± 0.000 | 0.006 ± 0.001 | 0.000 ± 0.000 |
| damage-visible | occluded-contact | 0.10   | 0.500 ± 0.104 | 0.297 ± 0.064 | 0.017 ± 0.002 | 0.000 ± 0.000 |
| visible-MPC    | occluded-contact | 0.10   | 0.500 ± 0.104 | 0.297 ± 0.064 | 0.017 ± 0.002 | 0.000 ± 0.000 |
| particle       | occluded-contact | 0.10   | 0.406 ± 0.101 | 0.328 ± 0.113 | 0.016 ± 0.001 | 0.047 ± 0.064 |
| robust-CEM     | occluded-contact | 0.10   | 0.406 ± 0.101 | 0.281 ± 0.101 | 0.016 ± 0.001 | 0.000 ± 0.000 |
| safe-particle  | occluded-contact | 0.10   | 0.406 ± 0.101 | 0.297 ± 0.079 | 0.016 ± 0.001 | 0.000 ± 0.000 |

Table 3 continued

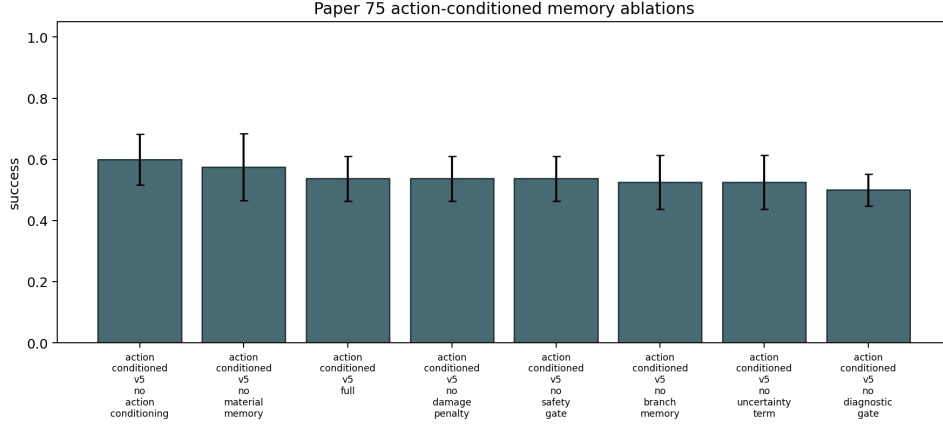


Figure 4: Ablation success for v5 components on combined memory stress.

| method         | split            | budget | success           | damage            | clip              | diag              |
|----------------|------------------|--------|-------------------|-------------------|-------------------|-------------------|
| ACM-v4         | occluded-contact | 0.10   | 0.391 $\pm$ 0.126 | 0.266 $\pm$ 0.098 | 0.013 $\pm$ 0.001 | 0.078 $\pm$ 0.064 |
| ACM-v5         | occluded-contact | 0.10   | 0.391 $\pm$ 0.108 | 0.281 $\pm$ 0.101 | 0.014 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| ensemble       | occluded-contact | 0.10   | 0.375 $\pm$ 0.131 | 0.234 $\pm$ 0.098 | 0.011 $\pm$ 0.001 | 0.094 $\pm$ 0.061 |
| oracle         | occluded-contact | 0.20   | 0.875 $\pm$ 0.080 | 0.000 $\pm$ 0.000 | 0.006 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| damage-visible | occluded-contact | 0.20   | 0.500 $\pm$ 0.104 | 0.297 $\pm$ 0.064 | 0.017 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| visible-MPC    | occluded-contact | 0.20   | 0.500 $\pm$ 0.104 | 0.297 $\pm$ 0.064 | 0.017 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| safe-particle  | occluded-contact | 0.20   | 0.406 $\pm$ 0.101 | 0.297 $\pm$ 0.079 | 0.015 $\pm$ 0.001 | 0.109 $\pm$ 0.072 |
| ACM-v5         | occluded-contact | 0.20   | 0.391 $\pm$ 0.108 | 0.281 $\pm$ 0.101 | 0.014 $\pm$ 0.001 | 0.016 $\pm$ 0.031 |
| robust-CEM     | occluded-contact | 0.20   | 0.391 $\pm$ 0.108 | 0.391 $\pm$ 0.072 | 0.013 $\pm$ 0.002 | 0.500 $\pm$ 0.093 |
| particle       | occluded-contact | 0.20   | 0.359 $\pm$ 0.117 | 0.516 $\pm$ 0.072 | 0.010 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| ACM-v4         | occluded-contact | 0.20   | 0.344 $\pm$ 0.111 | 0.516 $\pm$ 0.072 | 0.011 $\pm$ 0.001 | 0.922 $\pm$ 0.064 |
| ensemble       | occluded-contact | 0.20   | 0.344 $\pm$ 0.111 | 0.516 $\pm$ 0.072 | 0.010 $\pm$ 0.001 | 0.922 $\pm$ 0.064 |
| oracle         | occluded-contact | 0.30   | 0.875 $\pm$ 0.080 | 0.000 $\pm$ 0.000 | 0.006 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| damage-visible | occluded-contact | 0.30   | 0.500 $\pm$ 0.104 | 0.297 $\pm$ 0.064 | 0.017 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| visible-MPC    | occluded-contact | 0.30   | 0.500 $\pm$ 0.104 | 0.297 $\pm$ 0.064 | 0.017 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| safe-particle  | occluded-contact | 0.30   | 0.406 $\pm$ 0.101 | 0.297 $\pm$ 0.079 | 0.015 $\pm$ 0.001 | 0.109 $\pm$ 0.072 |
| ACM-v5         | occluded-contact | 0.30   | 0.391 $\pm$ 0.108 | 0.281 $\pm$ 0.101 | 0.014 $\pm$ 0.001 | 0.016 $\pm$ 0.031 |
| robust-CEM     | occluded-contact | 0.30   | 0.391 $\pm$ 0.108 | 0.391 $\pm$ 0.072 | 0.013 $\pm$ 0.002 | 0.500 $\pm$ 0.093 |
| particle       | occluded-contact | 0.30   | 0.359 $\pm$ 0.117 | 0.516 $\pm$ 0.072 | 0.010 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| ACM-v4         | occluded-contact | 0.30   | 0.344 $\pm$ 0.111 | 0.516 $\pm$ 0.072 | 0.011 $\pm$ 0.001 | 0.922 $\pm$ 0.064 |
| ensemble       | occluded-contact | 0.30   | 0.344 $\pm$ 0.111 | 0.516 $\pm$ 0.072 | 0.010 $\pm$ 0.001 | 0.922 $\pm$ 0.064 |
| oracle         | occluded-contact | 0.40   | 0.875 $\pm$ 0.080 | 0.000 $\pm$ 0.000 | 0.006 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| damage-visible | occluded-contact | 0.40   | 0.500 $\pm$ 0.104 | 0.297 $\pm$ 0.064 | 0.017 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| visible-MPC    | occluded-contact | 0.40   | 0.500 $\pm$ 0.104 | 0.297 $\pm$ 0.064 | 0.017 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| safe-particle  | occluded-contact | 0.40   | 0.406 $\pm$ 0.101 | 0.297 $\pm$ 0.079 | 0.015 $\pm$ 0.001 | 0.109 $\pm$ 0.072 |
| ACM-v5         | occluded-contact | 0.40   | 0.391 $\pm$ 0.108 | 0.281 $\pm$ 0.101 | 0.014 $\pm$ 0.001 | 0.016 $\pm$ 0.031 |

Table 3 continued

| method     | split            | budget | success           | damage            | clip              | diag              |
|------------|------------------|--------|-------------------|-------------------|-------------------|-------------------|
| robust-CEM | occluded-contact | 0.40   | 0.391 $\pm$ 0.108 | 0.391 $\pm$ 0.072 | 0.013 $\pm$ 0.002 | 0.500 $\pm$ 0.093 |
| particle   | occluded-contact | 0.40   | 0.359 $\pm$ 0.117 | 0.516 $\pm$ 0.072 | 0.010 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| ACM-v4     | occluded-contact | 0.40   | 0.344 $\pm$ 0.111 | 0.516 $\pm$ 0.072 | 0.011 $\pm$ 0.001 | 0.922 $\pm$ 0.064 |
| ensemble   | occluded-contact | 0.40   | 0.344 $\pm$ 0.111 | 0.516 $\pm$ 0.072 | 0.010 $\pm$ 0.001 | 0.922 $\pm$ 0.064 |

## 9 ABLATIONS

The ablations remove action conditioning, branch memory, safety gates, diagnostic gates, damage penalties, material memory, and uncertainty terms from the v5 planner. The ablation gate is intentionally strict: if an ablation matches full v5, the claimed mechanism is not necessary. Full v5 reaches  $0.537 \pm 0.073$  success.

Table 4: V5 ablations.

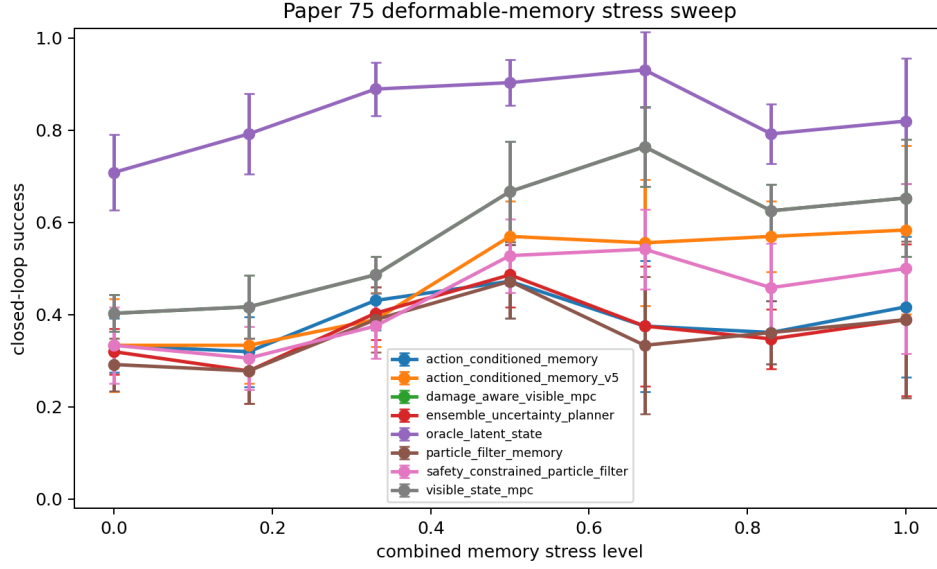


Figure 5: Stress sweep over combined hidden strain, occlusion, material shift, fixture strength, force limits, and observation noise.

| ablation       | success           | damage            | mem err           | mech F1           | diag              |
|----------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| no-action      | 0.600 $\pm$ 0.083 | 0.212 $\pm$ 0.094 | 0.523 $\pm$ 0.051 | 0.438 $\pm$ 0.065 | 0.000 $\pm$ 0.000 |
| no-material    | 0.575 $\pm$ 0.110 | 0.263 $\pm$ 0.111 | 0.237 $\pm$ 0.017 | 0.469 $\pm$ 0.077 | 0.000 $\pm$ 0.000 |
| full-v5        | 0.537 $\pm$ 0.073 | 0.275 $\pm$ 0.103 | 0.266 $\pm$ 0.028 | 0.453 $\pm$ 0.064 | 0.000 $\pm$ 0.000 |
| no-damage      | 0.537 $\pm$ 0.073 | 0.275 $\pm$ 0.103 | 0.266 $\pm$ 0.028 | 0.453 $\pm$ 0.064 | 0.000 $\pm$ 0.000 |
| no-safety      | 0.537 $\pm$ 0.073 | 0.275 $\pm$ 0.103 | 0.266 $\pm$ 0.028 | 0.453 $\pm$ 0.064 | 0.000 $\pm$ 0.000 |
| no-branch      | 0.525 $\pm$ 0.089 | 0.287 $\pm$ 0.078 | 0.266 $\pm$ 0.028 | 0.539 $\pm$ 0.069 | 0.000 $\pm$ 0.000 |
| no-uncertainty | 0.525 $\pm$ 0.089 | 0.287 $\pm$ 0.078 | 0.266 $\pm$ 0.028 | 0.539 $\pm$ 0.069 | 0.000 $\pm$ 0.000 |
| no-diag-gate   | 0.500 $\pm$ 0.052 | 0.287 $\pm$ 0.094 | 0.266 $\pm$ 0.028 | 0.453 $\pm$ 0.064 | 0.138 $\pm$ 0.063 |

## 10 STRESS SWEEP

At maximum stress level 1.00, ACM-v5 reaches  $0.583 \pm 0.183$  success, while the strongest non-oracle stress baseline, damage-visible, reaches  $0.653 \pm 0.126$ . This gate is reported even if it is unfavorable.

Table 5: Stress sweep metrics.



| method         | level | success           | damage            | clip              | diag              |
|----------------|-------|-------------------|-------------------|-------------------|-------------------|
| ACM-v4         | 0.00  | 0.333 $\pm$ 0.058 | 0.347 $\pm$ 0.076 | 0.001 $\pm$ 0.001 | 0.792 $\pm$ 0.104 |
| ACM-v5         | 0.00  | 0.333 $\pm$ 0.101 | 0.264 $\pm$ 0.091 | 0.007 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| damage-visible | 0.00  | 0.403 $\pm$ 0.040 | 0.264 $\pm$ 0.040 | 0.008 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| ensemble       | 0.00  | 0.319 $\pm$ 0.049 | 0.347 $\pm$ 0.076 | 0.002 $\pm$ 0.001 | 0.750 $\pm$ 0.090 |
| oracle         | 0.00  | 0.708 $\pm$ 0.082 | 0.000 $\pm$ 0.000 | 0.001 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| particle       | 0.00  | 0.292 $\pm$ 0.057 | 0.389 $\pm$ 0.058 | 0.000 $\pm$ 0.001 | 0.889 $\pm$ 0.058 |
| safe-particle  | 0.00  | 0.333 $\pm$ 0.082 | 0.292 $\pm$ 0.040 | 0.008 $\pm$ 0.002 | 0.028 $\pm$ 0.036 |
| visible-MPC    | 0.00  | 0.403 $\pm$ 0.040 | 0.264 $\pm$ 0.040 | 0.008 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| ACM-v4         | 0.17  | 0.319 $\pm$ 0.076 | 0.347 $\pm$ 0.087 | 0.006 $\pm$ 0.001 | 0.917 $\pm$ 0.054 |
| ACM-v5         | 0.17  | 0.333 $\pm$ 0.082 | 0.250 $\pm$ 0.068 | 0.011 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| damage-visible | 0.17  | 0.417 $\pm$ 0.068 | 0.250 $\pm$ 0.036 | 0.011 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| ensemble       | 0.17  | 0.278 $\pm$ 0.071 | 0.361 $\pm$ 0.080 | 0.006 $\pm$ 0.001 | 0.861 $\pm$ 0.099 |
| oracle         | 0.17  | 0.792 $\pm$ 0.087 | 0.014 $\pm$ 0.027 | 0.003 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| particle       | 0.17  | 0.278 $\pm$ 0.071 | 0.389 $\pm$ 0.058 | 0.005 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| safe-particle  | 0.17  | 0.306 $\pm$ 0.068 | 0.278 $\pm$ 0.058 | 0.010 $\pm$ 0.001 | 0.125 $\pm$ 0.076 |
| visible-MPC    | 0.17  | 0.417 $\pm$ 0.068 | 0.250 $\pm$ 0.036 | 0.011 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| ACM-v4         | 0.33  | 0.431 $\pm$ 0.064 | 0.417 $\pm$ 0.068 | 0.009 $\pm$ 0.002 | 0.917 $\pm$ 0.054 |
| ACM-v5         | 0.33  | 0.389 $\pm$ 0.058 | 0.250 $\pm$ 0.099 | 0.015 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| damage-visible | 0.33  | 0.486 $\pm$ 0.040 | 0.250 $\pm$ 0.036 | 0.016 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| ensemble       | 0.33  | 0.403 $\pm$ 0.057 | 0.403 $\pm$ 0.071 | 0.009 $\pm$ 0.002 | 0.875 $\pm$ 0.087 |
| oracle         | 0.33  | 0.889 $\pm$ 0.058 | 0.014 $\pm$ 0.027 | 0.006 $\pm$ 0.002 | 0.014 $\pm$ 0.027 |
| particle       | 0.33  | 0.389 $\pm$ 0.071 | 0.444 $\pm$ 0.058 | 0.009 $\pm$ 0.002 | 0.986 $\pm$ 0.027 |
| safe-particle  | 0.33  | 0.375 $\pm$ 0.071 | 0.292 $\pm$ 0.100 | 0.013 $\pm$ 0.002 | 0.250 $\pm$ 0.068 |
| visible-MPC    | 0.33  | 0.486 $\pm$ 0.040 | 0.250 $\pm$ 0.036 | 0.016 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| ACM-v4         | 0.50  | 0.472 $\pm$ 0.080 | 0.431 $\pm$ 0.064 | 0.019 $\pm$ 0.002 | 0.972 $\pm$ 0.036 |
| ACM-v5         | 0.50  | 0.569 $\pm$ 0.076 | 0.250 $\pm$ 0.080 | 0.022 $\pm$ 0.003 | 0.000 $\pm$ 0.000 |
| damage-visible | 0.50  | 0.667 $\pm$ 0.109 | 0.208 $\pm$ 0.064 | 0.024 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| ensemble       | 0.50  | 0.486 $\pm$ 0.071 | 0.417 $\pm$ 0.054 | 0.020 $\pm$ 0.001 | 0.944 $\pm$ 0.058 |

Table 5 continued

| method         | level | success           | damage            | clip              | diag              |
|----------------|-------|-------------------|-------------------|-------------------|-------------------|
| oracle         | 0.50  | 0.903 $\pm$ 0.049 | 0.028 $\pm$ 0.036 | 0.011 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| particle       | 0.50  | 0.472 $\pm$ 0.080 | 0.431 $\pm$ 0.064 | 0.020 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| safe-particle  | 0.50  | 0.528 $\pm$ 0.080 | 0.278 $\pm$ 0.071 | 0.021 $\pm$ 0.003 | 0.347 $\pm$ 0.096 |
| visible-MPC    | 0.50  | 0.667 $\pm$ 0.109 | 0.208 $\pm$ 0.064 | 0.024 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |
| ACM-v4         | 0.67  | 0.375 $\pm$ 0.142 | 0.556 $\pm$ 0.165 | 0.025 $\pm$ 0.002 | 0.917 $\pm$ 0.068 |
| ACM-v5         | 0.67  | 0.556 $\pm$ 0.137 | 0.264 $\pm$ 0.082 | 0.024 $\pm$ 0.003 | 0.000 $\pm$ 0.000 |
| damage-visible | 0.67  | 0.764 $\pm$ 0.087 | 0.139 $\pm$ 0.068 | 0.030 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| ensemble       | 0.67  | 0.375 $\pm$ 0.130 | 0.556 $\pm$ 0.148 | 0.025 $\pm$ 0.002 | 0.917 $\pm$ 0.054 |
| oracle         | 0.67  | 0.931 $\pm$ 0.082 | 0.014 $\pm$ 0.027 | 0.016 $\pm$ 0.004 | 0.000 $\pm$ 0.000 |
| particle       | 0.67  | 0.333 $\pm$ 0.148 | 0.625 $\pm$ 0.164 | 0.024 $\pm$ 0.002 | 0.986 $\pm$ 0.027 |
| safe-particle  | 0.67  | 0.542 $\pm$ 0.087 | 0.278 $\pm$ 0.082 | 0.026 $\pm$ 0.003 | 0.389 $\pm$ 0.130 |
| visible-MPC    | 0.67  | 0.764 $\pm$ 0.087 | 0.139 $\pm$ 0.068 | 0.030 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| ACM-v4         | 0.83  | 0.361 $\pm$ 0.068 | 0.597 $\pm$ 0.071 | 0.037 $\pm$ 0.002 | 0.958 $\pm$ 0.057 |
| ACM-v5         | 0.83  | 0.569 $\pm$ 0.076 | 0.292 $\pm$ 0.071 | 0.027 $\pm$ 0.004 | 0.000 $\pm$ 0.000 |
| damage-visible | 0.83  | 0.625 $\pm$ 0.057 | 0.264 $\pm$ 0.057 | 0.036 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| ensemble       | 0.83  | 0.347 $\pm$ 0.064 | 0.611 $\pm$ 0.071 | 0.036 $\pm$ 0.003 | 0.986 $\pm$ 0.027 |
| oracle         | 0.83  | 0.792 $\pm$ 0.064 | 0.083 $\pm$ 0.068 | 0.024 $\pm$ 0.003 | 0.028 $\pm$ 0.036 |
| particle       | 0.83  | 0.361 $\pm$ 0.068 | 0.597 $\pm$ 0.082 | 0.037 $\pm$ 0.002 | 0.972 $\pm$ 0.036 |
| safe-particle  | 0.83  | 0.458 $\pm$ 0.096 | 0.417 $\pm$ 0.122 | 0.033 $\pm$ 0.003 | 0.500 $\pm$ 0.123 |
| visible-MPC    | 0.83  | 0.625 $\pm$ 0.057 | 0.264 $\pm$ 0.057 | 0.036 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| ACM-v4         | 1.00  | 0.417 $\pm$ 0.153 | 0.542 $\pm$ 0.172 | 0.050 $\pm$ 0.003 | 0.972 $\pm$ 0.036 |
| ACM-v5         | 1.00  | 0.583 $\pm$ 0.183 | 0.333 $\pm$ 0.179 | 0.037 $\pm$ 0.005 | 0.000 $\pm$ 0.000 |
| damage-visible | 1.00  | 0.653 $\pm$ 0.126 | 0.236 $\pm$ 0.139 | 0.046 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |
| ensemble       | 1.00  | 0.389 $\pm$ 0.165 | 0.569 $\pm$ 0.186 | 0.049 $\pm$ 0.003 | 0.958 $\pm$ 0.040 |
| oracle         | 1.00  | 0.819 $\pm$ 0.136 | 0.083 $\pm$ 0.080 | 0.033 $\pm$ 0.005 | 0.028 $\pm$ 0.036 |
| particle       | 1.00  | 0.389 $\pm$ 0.170 | 0.569 $\pm$ 0.186 | 0.049 $\pm$ 0.003 | 0.958 $\pm$ 0.040 |
| safe-particle  | 1.00  | 0.500 $\pm$ 0.184 | 0.389 $\pm$ 0.210 | 0.045 $\pm$ 0.002 | 0.444 $\pm$ 0.082 |
| visible-MPC    | 1.00  | 0.653 $\pm$ 0.126 | 0.236 $\pm$ 0.139 | 0.046 $\pm$ 0.002 | 0.000 $\pm$ 0.000 |

## 11 NEGATIVE CASES

Negative cases are not anecdotes used to excuse the result; they are scenario-level checks that the average tables do not hide systematic failure modes. When a successful non-oracle baseline exists on the same scenario, the table lists it.

Table 6: Scenario-level negative cases for v5.

| seed | object           | v5 action          | successful baseline | baseline action        | failures             | shape |
|------|------------------|--------------------|---------------------|------------------------|----------------------|-------|
| 0    | rope             | fixture.avoid.pull |                     |                        | branch,shape         | 0.187 |
| 0    | cloth.strip      | gentle.relax.pull  |                     |                        | snag                 | 0.147 |
| 0    | deformable.sheet | fixture.avoid.pull |                     |                        | snag                 | 0.147 |
| 0    | rope             | fixture.avoid.pull |                     |                        | stretch,occ          | 0.171 |
| 0    | rope             | fixture.avoid.pull |                     |                        | branch,occ,shape     | 0.191 |
| 0    | cloth.strip      | fixture.avoid.pull |                     |                        | occ,shape            | 0.161 |
| 1    | rope             | fixture.avoid.pull |                     |                        | stretch              | 0.173 |
| 1    | cloth.strip      | fixture.avoid.pull | graph               | gentle.relax.pull      | shape                | 0.165 |
| 1    | rope             | fixture.avoid.pull |                     |                        | stretch,occ          | 0.171 |
| 1    | elastic.band     | gentle.relax.pull  | visible-MPC         | fixture.avoid.pull     | snag                 | 0.093 |
| 1    | rope             | fixture.avoid.pull |                     |                        | stretch,branch,shape | 0.196 |
| 1    | cloth.strip      | fixture.avoid.pull | history             | diagnostic.branch.pull | occ,shape            | 0.162 |

## 12 LIMITATIONS

This artifact is not a real-robot paper. It has no hardware videos, no public deformable benchmark, no tactile sensing, no RGBD perception stack, and no large neural dynamics model. It uses Ridge regressors from scikit-learn (Pedregosa et al., 2011) and a local simulator so that the evidence is cheap, reproducible, and easy to audit. Those constraints are deliberate, but they limit external validity. The right conclusion is not that deformable memory is useless; it is that this specific action-conditioned memory planner is not yet a submission-worthy control result.

## 13 CONCLUSION

The v5 rebuild improves the scientific artifact and sharpens the negative result. It adds the theory that v4 was missing, tries the obvious safety-gated rescue, evaluates stronger baselines, reports fixed-risk and stress behavior, and preserves failures. The frozen decision remains **KILL ARCHIVE**. A future revival would need either safer diagnostic actions, a richer low-risk action library, hardware validation, or a learned dynamics model that turns hidden memory into a consistent closed-loop advantage.

## A FULL MAIN METRICS

Table 7: All method/split main metrics.

| method         | split          | success           | damage            | mem err           | mech F1           | diag              |
|----------------|----------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| oracle         | combined       | 0.821 $\pm$ 0.084 | 0.062 $\pm$ 0.041 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.018 $\pm$ 0.023 |
| damage-visible | combined       | 0.589 $\pm$ 0.058 | 0.205 $\pm$ 0.041 | 0.578 $\pm$ 0.018 | 0.406 $\pm$ 0.040 | 0.000 $\pm$ 0.000 |
| visible-MPC    | combined       | 0.589 $\pm$ 0.058 | 0.205 $\pm$ 0.041 | 0.578 $\pm$ 0.018 | 0.406 $\pm$ 0.040 | 0.000 $\pm$ 0.000 |
| ACM-v5         | combined       | 0.562 $\pm$ 0.062 | 0.250 $\pm$ 0.075 | 0.229 $\pm$ 0.031 | 0.438 $\pm$ 0.065 | 0.000 $\pm$ 0.000 |
| graph          | combined       | 0.536 $\pm$ 0.092 | 0.259 $\pm$ 0.064 | 0.525 $\pm$ 0.032 | 0.427 $\pm$ 0.043 | 0.000 $\pm$ 0.000 |
| safe-particle  | combined       | 0.518 $\pm$ 0.063 | 0.330 $\pm$ 0.064 | 0.256 $\pm$ 0.031 | 0.391 $\pm$ 0.031 | 0.446 $\pm$ 0.069 |
| robust-CEM     | combined       | 0.482 $\pm$ 0.058 | 0.420 $\pm$ 0.041 | 0.255 $\pm$ 0.031 | 0.406 $\pm$ 0.040 | 0.786 $\pm$ 0.070 |
| ACM-v4         | combined       | 0.446 $\pm$ 0.063 | 0.482 $\pm$ 0.051 | 0.206 $\pm$ 0.032 | 0.453 $\pm$ 0.045 | 0.955 $\pm$ 0.026 |
| history        | combined       | 0.438 $\pm$ 0.062 | 0.482 $\pm$ 0.035 | 0.187 $\pm$ 0.029 | 0.438 $\pm$ 0.046 | 0.929 $\pm$ 0.037 |
| particle       | combined       | 0.438 $\pm$ 0.056 | 0.500 $\pm$ 0.046 | 0.187 $\pm$ 0.029 | 0.500 $\pm$ 0.000 | 0.964 $\pm$ 0.026 |
| ensemble       | combined       | 0.429 $\pm$ 0.059 | 0.500 $\pm$ 0.037 | 0.187 $\pm$ 0.030 | 0.469 $\pm$ 0.040 | 0.973 $\pm$ 0.026 |
| oracle         | hidden-strain  | 0.893 $\pm$ 0.046 | 0.027 $\pm$ 0.026 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| damage-visible | hidden-strain  | 0.536 $\pm$ 0.059 | 0.241 $\pm$ 0.037 | 0.480 $\pm$ 0.009 | 0.578 $\pm$ 0.045 | 0.000 $\pm$ 0.000 |
| visible-MPC    | hidden-strain  | 0.536 $\pm$ 0.059 | 0.241 $\pm$ 0.037 | 0.480 $\pm$ 0.009 | 0.578 $\pm$ 0.045 | 0.000 $\pm$ 0.000 |
| graph          | hidden-strain  | 0.473 $\pm$ 0.037 | 0.205 $\pm$ 0.062 | 0.382 $\pm$ 0.029 | 0.578 $\pm$ 0.045 | 0.000 $\pm$ 0.000 |
| history        | hidden-strain  | 0.446 $\pm$ 0.063 | 0.420 $\pm$ 0.049 | 0.180 $\pm$ 0.024 | 0.594 $\pm$ 0.061 | 0.884 $\pm$ 0.059 |
| ACM-v4         | hidden-strain  | 0.429 $\pm$ 0.065 | 0.438 $\pm$ 0.049 | 0.189 $\pm$ 0.028 | 0.609 $\pm$ 0.056 | 0.893 $\pm$ 0.046 |
| ensemble       | hidden-strain  | 0.429 $\pm$ 0.065 | 0.438 $\pm$ 0.041 | 0.183 $\pm$ 0.024 | 0.578 $\pm$ 0.045 | 0.911 $\pm$ 0.058 |
| safe-particle  | hidden-strain  | 0.420 $\pm$ 0.049 | 0.232 $\pm$ 0.044 | 0.209 $\pm$ 0.027 | 0.568 $\pm$ 0.051 | 0.909 $\pm$ 0.018 |
| robust-CEM     | hidden-strain  | 0.411 $\pm$ 0.074 | 0.330 $\pm$ 0.052 | 0.210 $\pm$ 0.027 | 0.594 $\pm$ 0.040 | 0.464 $\pm$ 0.106 |
| particle       | hidden-strain  | 0.402 $\pm$ 0.064 | 0.464 $\pm$ 0.046 | 0.180 $\pm$ 0.024 | 0.609 $\pm$ 0.031 | 0.902 $\pm$ 0.052 |
| ACM-v5         | hidden-strain  | 0.393 $\pm$ 0.053 | 0.241 $\pm$ 0.052 | 0.198 $\pm$ 0.028 | 0.552 $\pm$ 0.051 | 0.000 $\pm$ 0.000 |
| oracle         | material-shift | 0.920 $\pm$ 0.041 | 0.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.018 $\pm$ 0.023 |
| damage-visible | material-shift | 0.491 $\pm$ 0.049 | 0.214 $\pm$ 0.037 | 0.295 $\pm$ 0.004 | 0.625 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |

Table 7 continued

| method         | split            | success           | damage            | mem err           | mech F1           | diag              |
|----------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| visible-MPC    | material-shift   | 0.491 $\pm$ 0.049 | 0.214 $\pm$ 0.037 | 0.295 $\pm$ 0.004 | 0.625 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| graph          | material-shift   | 0.420 $\pm$ 0.077 | 0.205 $\pm$ 0.062 | 0.271 $\pm$ 0.030 | 0.594 $\pm$ 0.040 | 0.000 $\pm$ 0.000 |
| history        | material-shift   | 0.384 $\pm$ 0.079 | 0.438 $\pm$ 0.081 | 0.150 $\pm$ 0.016 | 0.516 $\pm$ 0.031 | 0.777 $\pm$ 0.081 |
| safe-particle  | material-shift   | 0.375 $\pm$ 0.069 | 0.241 $\pm$ 0.059 | 0.165 $\pm$ 0.016 | 0.562 $\pm$ 0.046 | 0.054 $\pm$ 0.044 |
| ACM-v4         | material-shift   | 0.366 $\pm$ 0.072 | 0.473 $\pm$ 0.083 | 0.153 $\pm$ 0.015 | 0.516 $\pm$ 0.031 | 0.812 $\pm$ 0.052 |
| ACM-v5         | material-shift   | 0.366 $\pm$ 0.072 | 0.214 $\pm$ 0.070 | 0.157 $\pm$ 0.016 | 0.578 $\pm$ 0.045 | 0.000 $\pm$ 0.000 |
| ensemble       | material-shift   | 0.366 $\pm$ 0.072 | 0.464 $\pm$ 0.088 | 0.152 $\pm$ 0.016 | 0.516 $\pm$ 0.031 | 0.777 $\pm$ 0.041 |
| robust-CEM     | material-shift   | 0.366 $\pm$ 0.072 | 0.304 $\pm$ 0.058 | 0.165 $\pm$ 0.016 | 0.516 $\pm$ 0.031 | 0.348 $\pm$ 0.072 |
| particle       | material-shift   | 0.348 $\pm$ 0.085 | 0.509 $\pm$ 0.097 | 0.150 $\pm$ 0.016 | 0.578 $\pm$ 0.045 | 0.955 $\pm$ 0.045 |
| oracle         | nominal          | 0.705 $\pm$ 0.062 | 0.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| damage-visible | nominal          | 0.393 $\pm$ 0.070 | 0.250 $\pm$ 0.026 | 0.124 $\pm$ 0.003 | 0.625 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| visible-MPC    | nominal          | 0.393 $\pm$ 0.070 | 0.250 $\pm$ 0.026 | 0.124 $\pm$ 0.003 | 0.625 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| graph          | nominal          | 0.384 $\pm$ 0.070 | 0.214 $\pm$ 0.046 | 0.109 $\pm$ 0.013 | 0.625 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| ACM-v4         | nominal          | 0.366 $\pm$ 0.077 | 0.259 $\pm$ 0.026 | 0.111 $\pm$ 0.022 | 0.609 $\pm$ 0.031 | 0.214 $\pm$ 0.065 |
| ACM-v5         | nominal          | 0.366 $\pm$ 0.067 | 0.232 $\pm$ 0.035 | 0.104 $\pm$ 0.021 | 0.625 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| safe-particle  | nominal          | 0.366 $\pm$ 0.067 | 0.250 $\pm$ 0.026 | 0.099 $\pm$ 0.018 | 0.625 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| robust-CEM     | nominal          | 0.357 $\pm$ 0.075 | 0.250 $\pm$ 0.026 | 0.099 $\pm$ 0.019 | 0.625 $\pm$ 0.000 | 0.045 $\pm$ 0.026 |
| ensemble       | nominal          | 0.348 $\pm$ 0.056 | 0.268 $\pm$ 0.023 | 0.128 $\pm$ 0.024 | 0.594 $\pm$ 0.040 | 0.339 $\pm$ 0.063 |
| history        | nominal          | 0.348 $\pm$ 0.056 | 0.259 $\pm$ 0.026 | 0.126 $\pm$ 0.021 | 0.609 $\pm$ 0.031 | 0.250 $\pm$ 0.059 |
| particle       | nominal          | 0.348 $\pm$ 0.049 | 0.295 $\pm$ 0.032 | 0.126 $\pm$ 0.021 | 0.547 $\pm$ 0.045 | 0.545 $\pm$ 0.091 |
| oracle         | occluded-contact | 0.830 $\pm$ 0.070 | 0.018 $\pm$ 0.023 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| damage-visible | occluded-contact | 0.536 $\pm$ 0.046 | 0.250 $\pm$ 0.026 | 0.393 $\pm$ 0.009 | 0.453 $\pm$ 0.045 | 0.000 $\pm$ 0.000 |
| visible-MPC    | occluded-contact | 0.536 $\pm$ 0.046 | 0.250 $\pm$ 0.026 | 0.393 $\pm$ 0.009 | 0.453 $\pm$ 0.045 | 0.000 $\pm$ 0.000 |
| safe-particle  | occluded-contact | 0.446 $\pm$ 0.035 | 0.295 $\pm$ 0.032 | 0.211 $\pm$ 0.026 | 0.406 $\pm$ 0.040 | 0.071 $\pm$ 0.053 |

Table 7 continued

| method     | split            | success           | damage            | mem err           | mech F1           | diag              |
|------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| ACM-v5     | occluded-contact | 0.420 $\pm$ 0.062 | 0.295 $\pm$ 0.041 | 0.203 $\pm$ 0.023 | 0.438 $\pm$ 0.046 | 0.000 $\pm$ 0.000 |
| graph      | occluded-contact | 0.420 $\pm$ 0.041 | 0.304 $\pm$ 0.035 | 0.357 $\pm$ 0.038 | 0.391 $\pm$ 0.031 | 0.000 $\pm$ 0.000 |
| robust-CEM | occluded-contact | 0.420 $\pm$ 0.049 | 0.339 $\pm$ 0.051 | 0.211 $\pm$ 0.026 | 0.406 $\pm$ 0.040 | 0.384 $\pm$ 0.083 |
| history    | occluded-contact | 0.393 $\pm$ 0.046 | 0.500 $\pm$ 0.059 | 0.197 $\pm$ 0.024 | 0.453 $\pm$ 0.045 | 0.839 $\pm$ 0.063 |
| ACM-v4     | occluded-contact | 0.384 $\pm$ 0.064 | 0.482 $\pm$ 0.044 | 0.199 $\pm$ 0.022 | 0.422 $\pm$ 0.045 | 0.795 $\pm$ 0.056 |
| ensemble   | occluded-contact | 0.384 $\pm$ 0.064 | 0.491 $\pm$ 0.041 | 0.200 $\pm$ 0.025 | 0.438 $\pm$ 0.046 | 0.812 $\pm$ 0.059 |
| particle   | occluded-contact | 0.357 $\pm$ 0.075 | 0.545 $\pm$ 0.064 | 0.197 $\pm$ 0.024 | 0.469 $\pm$ 0.040 | 0.929 $\pm$ 0.026 |

## B FULL PAIRWISE COMPARISONS

Table 8: Pairwise v5-minus-comparison statistics by split.

| split          | comparison     | d success | CI    | mem red | F1 diff | dmg red | better |
|----------------|----------------|-----------|-------|---------|---------|---------|--------|
| combined       | ACM-v4         | 0.116     | 0.045 | -0.023  | -0.016  | 0.232   | 7/8    |
| combined       | damage-visible | -0.027    | 0.052 | 0.349   | 0.031   | -0.045  | 2/8    |
| combined       | ensemble       | 0.134     | 0.056 | -0.042  | -0.031  | 0.250   | 7/8    |
| combined       | graph          | 0.027     | 0.087 | 0.296   | 0.010   | 0.009   | 4/8    |
| combined       | history        | 0.125     | 0.051 | -0.042  | 0.000   | 0.232   | 7/8    |
| combined       | oracle         | -0.259    | 0.083 | -0.229  | -0.562  | -0.188  | 0/8    |
| combined       | particle       | 0.125     | 0.063 | -0.042  | -0.062  | 0.250   | 6/8    |
| combined       | robust-CEM     | 0.080     | 0.049 | 0.026   | 0.031   | 0.170   | 6/8    |
| combined       | safe-particle  | 0.045     | 0.037 | 0.027   | 0.047   | 0.080   | 4/8    |
| combined       | visible-MPC    | -0.027    | 0.052 | 0.349   | 0.031   | -0.045  | 2/8    |
| hidden-strain  | ACM-v4         | -0.036    | 0.079 | -0.009  | -0.057  | 0.196   | 3/8    |
| hidden-strain  | damage-visible | -0.143    | 0.070 | 0.282   | -0.026  | 0.000   | 0/8    |
| hidden-strain  | ensemble       | -0.036    | 0.079 | -0.015  | -0.026  | 0.196   | 2/8    |
| hidden-strain  | graph          | -0.080    | 0.041 | 0.184   | -0.026  | -0.036  | 0/8    |
| hidden-strain  | history        | -0.054    | 0.074 | -0.018  | -0.042  | 0.179   | 2/8    |
| hidden-strain  | oracle         | -0.500    | 0.059 | -0.198  | -0.448  | -0.214  | 0/8    |
| hidden-strain  | particle       | -0.009    | 0.089 | -0.018  | -0.057  | 0.223   | 3/8    |
| hidden-strain  | robust-CEM     | -0.018    | 0.069 | 0.012   | -0.042  | 0.089   | 2/8    |
| hidden-strain  | safe-particle  | -0.027    | 0.026 | 0.011   | -0.016  | -0.009  | 0/8    |
| hidden-strain  | visible-MPC    | -0.143    | 0.070 | 0.282   | -0.026  | 0.000   | 0/8    |
| material-shift | ACM-v4         | 0.000     | 0.092 | -0.004  | 0.062   | 0.259   | 3/8    |
| material-shift | damage-visible | -0.125    | 0.051 | 0.138   | -0.047  | 0.000   | 0/8    |
| material-shift | ensemble       | 0.000     | 0.095 | -0.005  | 0.062   | 0.250   | 3/8    |
| material-shift | graph          | -0.054    | 0.087 | 0.114   | -0.016  | -0.009  | 2/8    |

Table 8 continued

| split            | comparison     | d success | CI    | mem red | F1 diff | dmg red | better |
|------------------|----------------|-----------|-------|---------|---------|---------|--------|
| material-shift   | history        | -0.018    | 0.098 | -0.007  | 0.062   | 0.223   | 2/8    |
| material-shift   | oracle         | -0.554    | 0.051 | -0.157  | -0.422  | -0.214  | 0/8    |
| material-shift   | particle       | 0.018     | 0.094 | -0.007  | 0.000   | 0.295   | 3/8    |
| material-shift   | robust-CEM     | 0.000     | 0.079 | 0.008   | 0.062   | 0.089   | 3/8    |
| material-shift   | safe-particle  | -0.009    | 0.049 | 0.008   | 0.016   | 0.027   | 3/8    |
| material-shift   | visible-MPC    | -0.125    | 0.051 | 0.138   | -0.047  | 0.000   | 0/8    |
| nominal          | ACM-v4         | -0.000    | 0.037 | 0.007   | 0.016   | 0.027   | 2/8    |
| nominal          | damage-visible | -0.027    | 0.026 | 0.020   | 0.000   | 0.018   | 0/8    |
| nominal          | ensemble       | 0.018     | 0.044 | 0.024   | 0.031   | 0.036   | 4/8    |
| nominal          | graph          | -0.018    | 0.023 | 0.005   | 0.000   | -0.018  | 0/8    |
| nominal          | history        | 0.018     | 0.035 | 0.022   | 0.016   | 0.027   | 3/8    |
| nominal          | oracle         | -0.339    | 0.035 | -0.104  | -0.375  | -0.232  | 0/8    |
| nominal          | particle       | 0.018     | 0.035 | 0.022   | 0.078   | 0.062   | 3/8    |
| nominal          | robust-CEM     | 0.009     | 0.018 | -0.005  | 0.000   | 0.018   | 1/8    |
| nominal          | safe-particle  | 0.000     | 0.000 | -0.005  | 0.000   | 0.018   | 0/8    |
| nominal          | visible-MPC    | -0.027    | 0.026 | 0.020   | 0.000   | 0.018   | 0/8    |
| occluded-contact | ACM-v4         | 0.036     | 0.088 | -0.004  | 0.016   | 0.188   | 5/8    |
| occluded-contact | damage-visible | -0.116    | 0.045 | 0.190   | -0.016  | -0.045  | 0/8    |
| occluded-contact | ensemble       | 0.036     | 0.088 | -0.003  | 0.000   | 0.196   | 5/8    |
| occluded-contact | graph          | 0.000     | 0.065 | 0.154   | 0.047   | 0.009   | 2/8    |
| occluded-contact | history        | 0.027     | 0.070 | -0.006  | -0.016  | 0.205   | 4/8    |
| occluded-contact | oracle         | -0.411    | 0.098 | -0.203  | -0.562  | -0.277  | 0/8    |
| occluded-contact | particle       | 0.062     | 0.077 | -0.006  | -0.031  | 0.250   | 5/8    |
| occluded-contact | robust-CEM     | 0.000     | 0.053 | 0.009   | 0.031   | 0.045   | 3/8    |

Table 8 continued

| split            | comparison    | d success | CI    | mem red | F1 diff | dmg red | better |
|------------------|---------------|-----------|-------|---------|---------|---------|--------|
| occluded-contact | safe-particle | -0.027    | 0.045 | 0.008   | 0.031   | -0.000  | 1/8    |
| occluded-contact | visible-MPC   | -0.116    | 0.045 | 0.190   | -0.016  | -0.045  | 0/8    |

## C FIXED-RISK PAIRWISE COMPARISONS

Table 9: Fixed-risk pairwise comparisons.

| split    | budget | comparison     | d success | CI    | dmg red | better |
|----------|--------|----------------|-----------|-------|---------|--------|
| combined | 0.10   | ACM-v4         | 0.078     | 0.092 | 0.078   | 3/8    |
| combined | 0.10   | damage-visible | 0.000     | 0.046 | 0.016   | 1/8    |
| combined | 0.10   | ensemble       | 0.094     | 0.101 | 0.078   | 5/8    |
| combined | 0.10   | oracle         | -0.203    | 0.079 | -0.109  | 0/8    |
| combined | 0.10   | particle       | 0.188     | 0.080 | 0.203   | 7/8    |
| combined | 0.10   | robust-CEM     | 0.031     | 0.061 | 0.047   | 1/8    |
| combined | 0.10   | safe-particle  | 0.078     | 0.092 | 0.078   | 3/8    |
| combined | 0.10   | visible-MPC    | 0.000     | 0.046 | 0.016   | 1/8    |
| combined | 0.20   | ACM-v4         | 0.312     | 0.179 | 0.469   | 7/8    |
| combined | 0.20   | damage-visible | 0.000     | 0.046 | 0.016   | 1/8    |
| combined | 0.20   | ensemble       | 0.297     | 0.173 | 0.453   | 7/8    |
| combined | 0.20   | oracle         | -0.203    | 0.079 | -0.109  | 0/8    |
| combined | 0.20   | particle       | 0.312     | 0.179 | 0.469   | 7/8    |
| combined | 0.20   | robust-CEM     | 0.188     | 0.131 | 0.297   | 6/8    |
| combined | 0.20   | safe-particle  | 0.094     | 0.111 | 0.141   | 4/8    |
| combined | 0.20   | visible-MPC    | 0.000     | 0.046 | 0.016   | 1/8    |
| combined | 0.30   | ACM-v4         | 0.312     | 0.179 | 0.469   | 7/8    |
| combined | 0.30   | damage-visible | 0.000     | 0.046 | 0.016   | 1/8    |
| combined | 0.30   | ensemble       | 0.297     | 0.173 | 0.453   | 7/8    |
| combined | 0.30   | oracle         | -0.203    | 0.079 | -0.109  | 0/8    |
| combined | 0.30   | particle       | 0.312     | 0.179 | 0.469   | 7/8    |
| combined | 0.30   | robust-CEM     | 0.188     | 0.131 | 0.297   | 6/8    |
| combined | 0.30   | safe-particle  | 0.094     | 0.111 | 0.141   | 4/8    |
| combined | 0.30   | visible-MPC    | 0.000     | 0.046 | 0.016   | 1/8    |

Table 9 continued

| split         | budget | comparison     | d success | CI    | dmg red | better |
|---------------|--------|----------------|-----------|-------|---------|--------|
| combined      | 0.40   | ACM-v4         | 0.312     | 0.179 | 0.469   | 7/8    |
| combined      | 0.40   | damage-visible | 0.000     | 0.046 | 0.016   | 1/8    |
| combined      | 0.40   | ensemble       | 0.297     | 0.173 | 0.453   | 7/8    |
| combined      | 0.40   | oracle         | -0.203    | 0.079 | -0.109  | 0/8    |
| combined      | 0.40   | particle       | 0.312     | 0.179 | 0.469   | 7/8    |
| combined      | 0.40   | robust-CEM     | 0.188     | 0.131 | 0.297   | 6/8    |
| combined      | 0.40   | safe-particle  | 0.094     | 0.111 | 0.141   | 4/8    |
| combined      | 0.40   | visible-MPC    | 0.000     | 0.046 | 0.016   | 1/8    |
| hidden-strain | 0.10   | ACM-v4         | 0.000     | 0.046 | 0.031   | 1/8    |
| hidden-strain | 0.10   | damage-visible | -0.062    | 0.046 | 0.094   | 0/8    |
| hidden-strain | 0.10   | ensemble       | 0.016     | 0.072 | 0.016   | 1/8    |
| hidden-strain | 0.10   | oracle         | -0.422    | 0.092 | -0.156  | 0/8    |
| hidden-strain | 0.10   | particle       | 0.000     | 0.046 | 0.094   | 1/8    |
| hidden-strain | 0.10   | robust-CEM     | -0.031    | 0.061 | 0.016   | 0/8    |
| hidden-strain | 0.10   | safe-particle  | -0.016    | 0.031 | 0.062   | 0/8    |
| hidden-strain | 0.10   | visible-MPC    | -0.062    | 0.046 | 0.094   | 0/8    |
| hidden-strain | 0.20   | ACM-v4         | 0.016     | 0.108 | 0.250   | 1/8    |
| hidden-strain | 0.20   | damage-visible | -0.062    | 0.046 | 0.094   | 0/8    |
| hidden-strain | 0.20   | ensemble       | 0.016     | 0.108 | 0.250   | 1/8    |
| hidden-strain | 0.20   | oracle         | -0.422    | 0.092 | -0.156  | 0/8    |
| hidden-strain | 0.20   | particle       | 0.031     | 0.111 | 0.266   | 2/8    |
| hidden-strain | 0.20   | robust-CEM     | 0.000     | 0.104 | 0.188   | 2/8    |
| hidden-strain | 0.20   | safe-particle  | -0.016    | 0.031 | 0.062   | 0/8    |
| hidden-strain | 0.20   | visible-MPC    | -0.062    | 0.046 | 0.094   | 0/8    |

Table 9 continued

| split            | budget | comparison     | d success | CI    | dmg red | better |
|------------------|--------|----------------|-----------|-------|---------|--------|
| hidden-strain    | 0.30   | ACM-v4         | 0.016     | 0.108 | 0.250   | 1/8    |
| hidden-strain    | 0.30   | damage-visible | -0.062    | 0.046 | 0.094   | 0/8    |
| hidden-strain    | 0.30   | ensemble       | 0.016     | 0.108 | 0.250   | 1/8    |
| hidden-strain    | 0.30   | oracle         | -0.422    | 0.092 | -0.156  | 0/8    |
| hidden-strain    | 0.30   | particle       | 0.031     | 0.111 | 0.266   | 2/8    |
| hidden-strain    | 0.30   | robust-CEM     | 0.000     | 0.104 | 0.188   | 2/8    |
| hidden-strain    | 0.30   | safe-particle  | -0.016    | 0.031 | 0.062   | 0/8    |
| hidden-strain    | 0.30   | visible-MPC    | -0.062    | 0.046 | 0.094   | 0/8    |
| hidden-strain    | 0.40   | ACM-v4         | 0.016     | 0.108 | 0.250   | 1/8    |
| hidden-strain    | 0.40   | damage-visible | -0.062    | 0.046 | 0.094   | 0/8    |
| hidden-strain    | 0.40   | ensemble       | 0.016     | 0.108 | 0.250   | 1/8    |
| hidden-strain    | 0.40   | oracle         | -0.422    | 0.092 | -0.156  | 0/8    |
| hidden-strain    | 0.40   | particle       | 0.031     | 0.111 | 0.266   | 2/8    |
| hidden-strain    | 0.40   | robust-CEM     | 0.000     | 0.104 | 0.188   | 2/8    |
| hidden-strain    | 0.40   | safe-particle  | -0.016    | 0.031 | 0.062   | 0/8    |
| hidden-strain    | 0.40   | visible-MPC    | -0.062    | 0.046 | 0.094   | 0/8    |
| occluded-contact | 0.10   | ACM-v4         | 0.000     | 0.046 | -0.016  | 1/8    |
| occluded-contact | 0.10   | damage-visible | -0.109    | 0.056 | 0.016   | 0/8    |
| occluded-contact | 0.10   | ensemble       | 0.016     | 0.056 | -0.047  | 2/8    |
| occluded-contact | 0.10   | oracle         | -0.484    | 0.098 | -0.281  | 0/8    |
| occluded-contact | 0.10   | particle       | -0.016    | 0.031 | 0.047   | 0/8    |
| occluded-contact | 0.10   | robust-CEM     | -0.016    | 0.031 | 0.000   | 0/8    |
| occluded-contact | 0.10   | safe-particle  | -0.016    | 0.031 | 0.016   | 0/8    |
| occluded-contact | 0.10   | visible-MPC    | -0.109    | 0.056 | 0.016   | 0/8    |

Table 9 continued

| split            | budget | comparison     | d success | CI    | dmg red | better |
|------------------|--------|----------------|-----------|-------|---------|--------|
| occluded-contact | 0.20   | ACM-v4         | 0.047     | 0.113 | 0.234   | 3/8    |
| occluded-contact | 0.20   | damage-visible | -0.109    | 0.056 | 0.016   | 0/8    |
| occluded-contact | 0.20   | ensemble       | 0.047     | 0.113 | 0.234   | 3/8    |
| occluded-contact | 0.20   | oracle         | -0.484    | 0.098 | -0.281  | 0/8    |
| occluded-contact | 0.20   | particle       | 0.031     | 0.137 | 0.234   | 3/8    |
| occluded-contact | 0.20   | robust-CEM     | 0.000     | 0.122 | 0.109   | 3/8    |
| occluded-contact | 0.20   | safe-particle  | -0.016    | 0.031 | 0.016   | 0/8    |
| occluded-contact | 0.20   | visible-MPC    | -0.109    | 0.056 | 0.016   | 0/8    |
| occluded-contact | 0.30   | ACM-v4         | 0.047     | 0.113 | 0.234   | 3/8    |
| occluded-contact | 0.30   | damage-visible | -0.109    | 0.056 | 0.016   | 0/8    |
| occluded-contact | 0.30   | ensemble       | 0.047     | 0.113 | 0.234   | 3/8    |
| occluded-contact | 0.30   | oracle         | -0.484    | 0.098 | -0.281  | 0/8    |
| occluded-contact | 0.30   | particle       | 0.031     | 0.137 | 0.234   | 3/8    |
| occluded-contact | 0.30   | robust-CEM     | 0.000     | 0.122 | 0.109   | 3/8    |
| occluded-contact | 0.30   | safe-particle  | -0.016    | 0.031 | 0.016   | 0/8    |
| occluded-contact | 0.30   | visible-MPC    | -0.109    | 0.056 | 0.016   | 0/8    |
| occluded-contact | 0.40   | ACM-v4         | 0.047     | 0.113 | 0.234   | 3/8    |
| occluded-contact | 0.40   | damage-visible | -0.109    | 0.056 | 0.016   | 0/8    |
| occluded-contact | 0.40   | ensemble       | 0.047     | 0.113 | 0.234   | 3/8    |
| occluded-contact | 0.40   | oracle         | -0.484    | 0.098 | -0.281  | 0/8    |
| occluded-contact | 0.40   | particle       | 0.031     | 0.137 | 0.234   | 3/8    |
| occluded-contact | 0.40   | robust-CEM     | 0.000     | 0.122 | 0.109   | 3/8    |
| occluded-contact | 0.40   | safe-particle  | -0.016    | 0.031 | 0.016   | 0/8    |
| occluded-contact | 0.40   | visible-MPC    | -0.109    | 0.056 | 0.016   | 0/8    |

## D PER-SEED MAIN METRICS

Paired statistics are only meaningful if the seed-level evidence is available. Table 10 therefore includes the per-seed, per-method, per-split summaries used to compute the main confidence intervals and pairwise differences.

Table 10: Per-seed main metrics.

| method         | split    | seed | n  | success | damage | mem err | F1    | diag  |
|----------------|----------|------|----|---------|--------|---------|-------|-------|
| ACM-v4         | combined | 0    | 14 | 0.357   | 0.571  | 0.230   | 0.500 | 1.000 |
| ACM-v4         | combined | 1    | 14 | 0.429   | 0.571  | 0.237   | 0.375 | 0.929 |
| ACM-v4         | combined | 2    | 14 | 0.571   | 0.357  | 0.142   | 0.500 | 0.929 |
| ACM-v4         | combined | 3    | 14 | 0.500   | 0.429  | 0.231   | 0.500 | 1.000 |
| ACM-v4         | combined | 4    | 14 | 0.286   | 0.500  | 0.186   | 0.375 | 0.929 |
| ACM-v4         | combined | 5    | 14 | 0.500   | 0.429  | 0.254   | 0.375 | 0.929 |
| ACM-v4         | combined | 6    | 14 | 0.500   | 0.500  | 0.135   | 0.500 | 0.929 |
| ACM-v4         | combined | 7    | 14 | 0.429   | 0.500  | 0.232   | 0.500 | 1.000 |
| ACM-v5         | combined | 0    | 14 | 0.571   | 0.214  | 0.257   | 0.375 | 0.000 |
| ACM-v5         | combined | 1    | 14 | 0.571   | 0.286  | 0.251   | 0.375 | 0.000 |
| ACM-v5         | combined | 2    | 14 | 0.714   | 0.071  | 0.164   | 0.625 | 0.000 |
| ACM-v5         | combined | 3    | 14 | 0.571   | 0.357  | 0.255   | 0.500 | 0.000 |
| ACM-v5         | combined | 4    | 14 | 0.429   | 0.357  | 0.216   | 0.375 | 0.000 |
| ACM-v5         | combined | 5    | 14 | 0.500   | 0.214  | 0.269   | 0.375 | 0.000 |
| ACM-v5         | combined | 6    | 14 | 0.643   | 0.143  | 0.157   | 0.500 | 0.000 |
| ACM-v5         | combined | 7    | 14 | 0.500   | 0.357  | 0.263   | 0.375 | 0.000 |
| damage-visible | combined | 0    | 14 | 0.500   | 0.143  | 0.578   | 0.375 | 0.000 |
| damage-visible | combined | 1    | 14 | 0.643   | 0.214  | 0.571   | 0.500 | 0.000 |
| damage-visible | combined | 2    | 14 | 0.714   | 0.143  | 0.588   | 0.375 | 0.000 |
| damage-visible | combined | 3    | 14 | 0.500   | 0.286  | 0.591   | 0.375 | 0.000 |
| damage-visible | combined | 4    | 14 | 0.500   | 0.286  | 0.617   | 0.375 | 0.000 |
| damage-visible | combined | 5    | 14 | 0.571   | 0.214  | 0.536   | 0.375 | 0.000 |
| damage-visible | combined | 6    | 14 | 0.643   | 0.214  | 0.594   | 0.375 | 0.000 |
| damage-visible | combined | 7    | 14 | 0.643   | 0.143  | 0.550   | 0.500 | 0.000 |
| ensemble       | combined | 0    | 14 | 0.357   | 0.571  | 0.202   | 0.500 | 1.000 |
| ensemble       | combined | 1    | 14 | 0.500   | 0.500  | 0.219   | 0.500 | 0.929 |
| ensemble       | combined | 2    | 14 | 0.500   | 0.429  | 0.129   | 0.500 | 1.000 |
| ensemble       | combined | 3    | 14 | 0.500   | 0.429  | 0.242   | 0.500 | 1.000 |
| ensemble       | combined | 4    | 14 | 0.286   | 0.500  | 0.163   | 0.375 | 0.929 |
| ensemble       | combined | 5    | 14 | 0.500   | 0.500  | 0.229   | 0.500 | 1.000 |

Table 10 continued

| method   | split    | seed | n  | success | damage | mem err | F1    | diag  |
|----------|----------|------|----|---------|--------|---------|-------|-------|
| ensemble | combined | 6    | 14 | 0.429   | 0.571  | 0.129   | 0.500 | 1.000 |
| ensemble | combined | 7    | 14 | 0.357   | 0.500  | 0.182   | 0.375 | 0.929 |
| graph    | combined | 0    | 14 | 0.429   | 0.214  | 0.537   | 0.500 | 0.000 |
| graph    | combined | 1    | 14 | 0.714   | 0.214  | 0.529   | 0.500 | 0.000 |
| graph    | combined | 2    | 14 | 0.714   | 0.214  | 0.509   | 0.375 | 0.000 |
| graph    | combined | 3    | 14 | 0.357   | 0.429  | 0.545   | 0.375 | 0.000 |
| graph    | combined | 4    | 14 | 0.571   | 0.286  | 0.583   | 0.375 | 0.000 |
| graph    | combined | 5    | 14 | 0.500   | 0.214  | 0.430   | 0.417 | 0.000 |
| graph    | combined | 6    | 14 | 0.571   | 0.143  | 0.506   | 0.500 | 0.000 |
| graph    | combined | 7    | 14 | 0.429   | 0.357  | 0.560   | 0.375 | 0.000 |
| history  | combined | 0    | 14 | 0.357   | 0.571  | 0.198   | 0.500 | 0.929 |
| history  | combined | 1    | 14 | 0.500   | 0.500  | 0.233   | 0.375 | 0.857 |
| history  | combined | 2    | 14 | 0.500   | 0.429  | 0.115   | 0.500 | 1.000 |
| history  | combined | 3    | 14 | 0.500   | 0.429  | 0.218   | 0.500 | 1.000 |
| history  | combined | 4    | 14 | 0.286   | 0.500  | 0.179   | 0.375 | 0.929 |
| history  | combined | 5    | 14 | 0.500   | 0.429  | 0.229   | 0.375 | 0.929 |
| history  | combined | 6    | 14 | 0.500   | 0.500  | 0.137   | 0.500 | 0.929 |
| history  | combined | 7    | 14 | 0.357   | 0.500  | 0.186   | 0.375 | 0.857 |
| oracle   | combined | 0    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.071 |
| oracle   | combined | 1    | 14 | 0.929   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | combined | 2    | 14 | 0.929   | 0.071  | 0.000   | 1.000 | 0.000 |
| oracle   | combined | 3    | 14 | 0.643   | 0.071  | 0.000   | 1.000 | 0.000 |
| oracle   | combined | 4    | 14 | 0.786   | 0.143  | 0.000   | 1.000 | 0.071 |
| oracle   | combined | 5    | 14 | 0.643   | 0.143  | 0.000   | 1.000 | 0.000 |
| oracle   | combined | 6    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | combined | 7    | 14 | 0.929   | 0.071  | 0.000   | 1.000 | 0.000 |
| particle | combined | 0    | 14 | 0.357   | 0.571  | 0.198   | 0.500 | 1.000 |
| particle | combined | 1    | 14 | 0.429   | 0.500  | 0.233   | 0.500 | 0.929 |
| particle | combined | 2    | 14 | 0.500   | 0.429  | 0.115   | 0.500 | 0.929 |
| particle | combined | 3    | 14 | 0.500   | 0.429  | 0.218   | 0.500 | 0.929 |

Table 10 continued

| method        | split         | seed | n  | success | damage | mem err | F1    | diag  |
|---------------|---------------|------|----|---------|--------|---------|-------|-------|
| particle      | combined      | 4    | 14 | 0.286   | 0.571  | 0.179   | 0.500 | 1.000 |
| particle      | combined      | 5    | 14 | 0.500   | 0.500  | 0.229   | 0.500 | 1.000 |
| particle      | combined      | 6    | 14 | 0.429   | 0.571  | 0.137   | 0.500 | 1.000 |
| particle      | combined      | 7    | 14 | 0.500   | 0.429  | 0.186   | 0.500 | 0.929 |
| robust-CEM    | combined      | 0    | 14 | 0.357   | 0.500  | 0.286   | 0.375 | 0.786 |
| robust-CEM    | combined      | 1    | 14 | 0.500   | 0.500  | 0.278   | 0.375 | 0.786 |
| robust-CEM    | combined      | 2    | 14 | 0.571   | 0.357  | 0.188   | 0.500 | 0.857 |
| robust-CEM    | combined      | 3    | 14 | 0.500   | 0.429  | 0.284   | 0.500 | 0.929 |
| robust-CEM    | combined      | 4    | 14 | 0.357   | 0.429  | 0.246   | 0.375 | 0.786 |
| robust-CEM    | combined      | 5    | 14 | 0.500   | 0.429  | 0.285   | 0.375 | 0.786 |
| robust-CEM    | combined      | 6    | 14 | 0.571   | 0.357  | 0.185   | 0.375 | 0.786 |
| robust-CEM    | combined      | 7    | 14 | 0.500   | 0.357  | 0.293   | 0.375 | 0.571 |
| safe-particle | combined      | 0    | 14 | 0.429   | 0.357  | 0.286   | 0.375 | 0.429 |
| safe-particle | combined      | 1    | 14 | 0.571   | 0.286  | 0.279   | 0.375 | 0.357 |
| safe-particle | combined      | 2    | 14 | 0.643   | 0.214  | 0.187   | 0.375 | 0.571 |
| safe-particle | combined      | 3    | 14 | 0.571   | 0.357  | 0.285   | 0.500 | 0.500 |
| safe-particle | combined      | 4    | 14 | 0.357   | 0.500  | 0.243   | 0.375 | 0.429 |
| safe-particle | combined      | 5    | 14 | 0.500   | 0.357  | 0.286   | 0.375 | 0.571 |
| safe-particle | combined      | 6    | 14 | 0.571   | 0.214  | 0.186   | 0.375 | 0.286 |
| safe-particle | combined      | 7    | 14 | 0.500   | 0.357  | 0.293   | 0.375 | 0.429 |
| visible-MPC   | combined      | 0    | 14 | 0.500   | 0.143  | 0.578   | 0.375 | 0.000 |
| visible-MPC   | combined      | 1    | 14 | 0.643   | 0.214  | 0.571   | 0.500 | 0.000 |
| visible-MPC   | combined      | 2    | 14 | 0.714   | 0.143  | 0.588   | 0.375 | 0.000 |
| visible-MPC   | combined      | 3    | 14 | 0.500   | 0.286  | 0.591   | 0.375 | 0.000 |
| visible-MPC   | combined      | 4    | 14 | 0.500   | 0.286  | 0.617   | 0.375 | 0.000 |
| visible-MPC   | combined      | 5    | 14 | 0.571   | 0.214  | 0.536   | 0.375 | 0.000 |
| visible-MPC   | combined      | 6    | 14 | 0.643   | 0.214  | 0.594   | 0.375 | 0.000 |
| visible-MPC   | combined      | 7    | 14 | 0.643   | 0.143  | 0.550   | 0.500 | 0.000 |
| ACM-v4        | hidden-strain | 0    | 14 | 0.571   | 0.286  | 0.152   | 0.750 | 0.857 |
| ACM-v4        | hidden-strain | 1    | 14 | 0.357   | 0.500  | 0.231   | 0.500 | 0.857 |

Table 10 continued

| method         | split         | seed | n  | success | damage | mem err | F1    | diag  |
|----------------|---------------|------|----|---------|--------|---------|-------|-------|
| ACM-v4         | hidden-strain | 2    | 14 | 0.500   | 0.500  | 0.139   | 0.625 | 0.929 |
| ACM-v4         | hidden-strain | 3    | 14 | 0.286   | 0.429  | 0.151   | 0.625 | 0.929 |
| ACM-v4         | hidden-strain | 4    | 14 | 0.500   | 0.429  | 0.175   | 0.625 | 0.929 |
| ACM-v4         | hidden-strain | 5    | 14 | 0.429   | 0.500  | 0.213   | 0.625 | 1.000 |
| ACM-v4         | hidden-strain | 6    | 14 | 0.357   | 0.429  | 0.245   | 0.625 | 0.857 |
| ACM-v4         | hidden-strain | 7    | 14 | 0.429   | 0.429  | 0.206   | 0.500 | 0.786 |
| ACM-v5         | hidden-strain | 0    | 14 | 0.429   | 0.143  | 0.165   | 0.625 | 0.000 |
| ACM-v5         | hidden-strain | 1    | 14 | 0.429   | 0.214  | 0.231   | 0.667 | 0.000 |
| ACM-v5         | hidden-strain | 2    | 14 | 0.357   | 0.286  | 0.152   | 0.500 | 0.000 |
| ACM-v5         | hidden-strain | 3    | 14 | 0.286   | 0.286  | 0.159   | 0.500 | 0.000 |
| ACM-v5         | hidden-strain | 4    | 14 | 0.286   | 0.357  | 0.177   | 0.500 | 0.000 |
| ACM-v5         | hidden-strain | 5    | 14 | 0.500   | 0.214  | 0.222   | 0.500 | 0.000 |
| ACM-v5         | hidden-strain | 6    | 14 | 0.429   | 0.286  | 0.262   | 0.500 | 0.000 |
| ACM-v5         | hidden-strain | 7    | 14 | 0.429   | 0.143  | 0.216   | 0.625 | 0.000 |
| damage-visible | hidden-strain | 0    | 14 | 0.500   | 0.214  | 0.473   | 0.625 | 0.000 |
| damage-visible | hidden-strain | 1    | 14 | 0.571   | 0.214  | 0.498   | 0.625 | 0.000 |
| damage-visible | hidden-strain | 2    | 14 | 0.714   | 0.214  | 0.459   | 0.500 | 0.000 |
| damage-visible | hidden-strain | 3    | 14 | 0.500   | 0.286  | 0.475   | 0.500 | 0.000 |
| damage-visible | hidden-strain | 4    | 14 | 0.429   | 0.286  | 0.479   | 0.625 | 0.000 |
| damage-visible | hidden-strain | 5    | 14 | 0.571   | 0.143  | 0.499   | 0.625 | 0.000 |
| damage-visible | hidden-strain | 6    | 14 | 0.500   | 0.286  | 0.484   | 0.500 | 0.000 |
| damage-visible | hidden-strain | 7    | 14 | 0.500   | 0.286  | 0.474   | 0.625 | 0.000 |
| ensemble       | hidden-strain | 0    | 14 | 0.500   | 0.357  | 0.156   | 0.625 | 1.000 |
| ensemble       | hidden-strain | 1    | 14 | 0.429   | 0.429  | 0.236   | 0.500 | 0.786 |
| ensemble       | hidden-strain | 2    | 14 | 0.500   | 0.500  | 0.129   | 0.625 | 0.929 |
| ensemble       | hidden-strain | 3    | 14 | 0.286   | 0.429  | 0.163   | 0.625 | 0.929 |
| ensemble       | hidden-strain | 4    | 14 | 0.500   | 0.429  | 0.170   | 0.625 | 0.929 |
| ensemble       | hidden-strain | 5    | 14 | 0.429   | 0.500  | 0.194   | 0.625 | 1.000 |
| ensemble       | hidden-strain | 6    | 14 | 0.286   | 0.500  | 0.216   | 0.500 | 0.929 |
| ensemble       | hidden-strain | 7    | 14 | 0.500   | 0.357  | 0.201   | 0.500 | 0.786 |

Table 10 continued



| method   | split         | seed | n  | success | damage | mem err | F1    | diag  |
|----------|---------------|------|----|---------|--------|---------|-------|-------|
| graph    | hidden-strain | 0    | 14 | 0.429   | 0.214  | 0.425   | 0.625 | 0.000 |
| graph    | hidden-strain | 1    | 14 | 0.429   | 0.071  | 0.369   | 0.625 | 0.000 |
| graph    | hidden-strain | 2    | 14 | 0.500   | 0.214  | 0.348   | 0.500 | 0.000 |
| graph    | hidden-strain | 3    | 14 | 0.429   | 0.357  | 0.356   | 0.500 | 0.000 |
| graph    | hidden-strain | 4    | 14 | 0.429   | 0.214  | 0.342   | 0.625 | 0.000 |
| graph    | hidden-strain | 5    | 14 | 0.571   | 0.143  | 0.388   | 0.625 | 0.000 |
| graph    | hidden-strain | 6    | 14 | 0.500   | 0.286  | 0.461   | 0.500 | 0.000 |
| graph    | hidden-strain | 7    | 14 | 0.500   | 0.143  | 0.366   | 0.625 | 0.000 |
| history  | hidden-strain | 0    | 14 | 0.571   | 0.286  | 0.151   | 0.750 | 0.929 |
| history  | hidden-strain | 1    | 14 | 0.429   | 0.429  | 0.225   | 0.500 | 0.786 |
| history  | hidden-strain | 2    | 14 | 0.500   | 0.500  | 0.127   | 0.625 | 0.929 |
| history  | hidden-strain | 3    | 14 | 0.286   | 0.429  | 0.148   | 0.625 | 0.929 |
| history  | hidden-strain | 4    | 14 | 0.500   | 0.429  | 0.174   | 0.625 | 0.929 |
| history  | hidden-strain | 5    | 14 | 0.429   | 0.500  | 0.196   | 0.625 | 1.000 |
| history  | hidden-strain | 6    | 14 | 0.357   | 0.429  | 0.217   | 0.500 | 0.786 |
| history  | hidden-strain | 7    | 14 | 0.500   | 0.357  | 0.199   | 0.500 | 0.786 |
| oracle   | hidden-strain | 0    | 14 | 1.000   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | hidden-strain | 1    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | hidden-strain | 2    | 14 | 0.929   | 0.071  | 0.000   | 1.000 | 0.000 |
| oracle   | hidden-strain | 3    | 14 | 0.929   | 0.071  | 0.000   | 1.000 | 0.000 |
| oracle   | hidden-strain | 4    | 14 | 0.786   | 0.071  | 0.000   | 1.000 | 0.000 |
| oracle   | hidden-strain | 5    | 14 | 0.929   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | hidden-strain | 6    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | hidden-strain | 7    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.000 |
| particle | hidden-strain | 0    | 14 | 0.500   | 0.357  | 0.151   | 0.625 | 1.000 |
| particle | hidden-strain | 1    | 14 | 0.357   | 0.571  | 0.225   | 0.625 | 1.000 |
| particle | hidden-strain | 2    | 14 | 0.500   | 0.500  | 0.127   | 0.625 | 0.929 |
| particle | hidden-strain | 3    | 14 | 0.286   | 0.429  | 0.148   | 0.625 | 0.857 |
| particle | hidden-strain | 4    | 14 | 0.500   | 0.429  | 0.174   | 0.625 | 0.857 |
| particle | hidden-strain | 5    | 14 | 0.357   | 0.500  | 0.196   | 0.625 | 0.857 |

Table 10 continued

| method        | split          | seed | n  | success | damage | mem err | F1    | diag  |
|---------------|----------------|------|----|---------|--------|---------|-------|-------|
| particle      | hidden-strain  | 6    | 14 | 0.286   | 0.500  | 0.217   | 0.625 | 0.929 |
| particle      | hidden-strain  | 7    | 14 | 0.429   | 0.429  | 0.199   | 0.500 | 0.786 |
| robust-CEM    | hidden-strain  | 0    | 14 | 0.571   | 0.214  | 0.189   | 0.625 | 0.571 |
| robust-CEM    | hidden-strain  | 1    | 14 | 0.429   | 0.286  | 0.235   | 0.500 | 0.500 |
| robust-CEM    | hidden-strain  | 2    | 14 | 0.357   | 0.357  | 0.166   | 0.625 | 0.500 |
| robust-CEM    | hidden-strain  | 3    | 14 | 0.214   | 0.357  | 0.169   | 0.625 | 0.500 |
| robust-CEM    | hidden-strain  | 4    | 14 | 0.429   | 0.429  | 0.182   | 0.625 | 0.643 |
| robust-CEM    | hidden-strain  | 5    | 14 | 0.357   | 0.429  | 0.231   | 0.625 | 0.500 |
| robust-CEM    | hidden-strain  | 6    | 14 | 0.429   | 0.286  | 0.279   | 0.500 | 0.143 |
| robust-CEM    | hidden-strain  | 7    | 14 | 0.500   | 0.286  | 0.227   | 0.625 | 0.357 |
| safe-particle | hidden-strain  | 0    | 14 | 0.500   | 0.214  | 0.190   | 0.625 | 0.000 |
| safe-particle | hidden-strain  | 1    | 14 | 0.429   | 0.214  | 0.228   | 0.667 | 0.000 |
| safe-particle | hidden-strain  | 2    | 14 | 0.429   | 0.214  | 0.166   | 0.500 | 0.000 |
| safe-particle | hidden-strain  | 3    | 14 | 0.286   | 0.357  | 0.171   | 0.500 | 0.000 |
| safe-particle | hidden-strain  | 4    | 14 | 0.357   | 0.214  | 0.182   | 0.625 | 0.000 |
| safe-particle | hidden-strain  | 5    | 14 | 0.500   | 0.214  | 0.230   | 0.500 | 0.071 |
| safe-particle | hidden-strain  | 6    | 14 | 0.429   | 0.286  | 0.280   | 0.500 | 0.000 |
| safe-particle | hidden-strain  | 7    | 14 | 0.429   | 0.143  | 0.227   | 0.625 | 0.000 |
| visible-MPC   | hidden-strain  | 0    | 14 | 0.500   | 0.214  | 0.473   | 0.625 | 0.000 |
| visible-MPC   | hidden-strain  | 1    | 14 | 0.571   | 0.214  | 0.498   | 0.625 | 0.000 |
| visible-MPC   | hidden-strain  | 2    | 14 | 0.714   | 0.214  | 0.459   | 0.500 | 0.000 |
| visible-MPC   | hidden-strain  | 3    | 14 | 0.500   | 0.286  | 0.475   | 0.500 | 0.000 |
| visible-MPC   | hidden-strain  | 4    | 14 | 0.429   | 0.286  | 0.479   | 0.625 | 0.000 |
| visible-MPC   | hidden-strain  | 5    | 14 | 0.571   | 0.143  | 0.499   | 0.625 | 0.000 |
| visible-MPC   | hidden-strain  | 6    | 14 | 0.500   | 0.286  | 0.484   | 0.500 | 0.000 |
| visible-MPC   | hidden-strain  | 7    | 14 | 0.500   | 0.286  | 0.474   | 0.625 | 0.000 |
| ACM-v4        | material-shift | 0    | 14 | 0.286   | 0.571  | 0.145   | 0.500 | 0.714 |
| ACM-v4        | material-shift | 1    | 14 | 0.286   | 0.357  | 0.141   | 0.500 | 0.786 |
| ACM-v4        | material-shift | 2    | 14 | 0.357   | 0.500  | 0.128   | 0.500 | 0.857 |
| ACM-v4        | material-shift | 3    | 14 | 0.500   | 0.286  | 0.190   | 0.500 | 0.714 |

Table 10 continued

| method         | split          | seed | n  | success | damage | mem err | F1    | diag  |
|----------------|----------------|------|----|---------|--------|---------|-------|-------|
| ACM-v4         | material-shift | 4    | 14 | 0.214   | 0.643  | 0.153   | 0.500 | 0.857 |
| ACM-v4         | material-shift | 5    | 14 | 0.429   | 0.429  | 0.158   | 0.500 | 0.929 |
| ACM-v4         | material-shift | 6    | 14 | 0.357   | 0.571  | 0.179   | 0.500 | 0.786 |
| ACM-v4         | material-shift | 7    | 14 | 0.500   | 0.429  | 0.131   | 0.625 | 0.857 |
| ACM-v5         | material-shift | 0    | 14 | 0.214   | 0.214  | 0.152   | 0.500 | 0.000 |
| ACM-v5         | material-shift | 1    | 14 | 0.357   | 0.214  | 0.149   | 0.625 | 0.000 |
| ACM-v5         | material-shift | 2    | 14 | 0.571   | 0.143  | 0.122   | 0.625 | 0.000 |
| ACM-v5         | material-shift | 3    | 14 | 0.357   | 0.071  | 0.192   | 0.625 | 0.000 |
| ACM-v5         | material-shift | 4    | 14 | 0.357   | 0.214  | 0.162   | 0.625 | 0.000 |
| ACM-v5         | material-shift | 5    | 14 | 0.286   | 0.143  | 0.166   | 0.625 | 0.000 |
| ACM-v5         | material-shift | 6    | 14 | 0.357   | 0.357  | 0.177   | 0.500 | 0.000 |
| ACM-v5         | material-shift | 7    | 14 | 0.429   | 0.357  | 0.135   | 0.500 | 0.000 |
| damage-visible | material-shift | 0    | 14 | 0.429   | 0.286  | 0.295   | 0.625 | 0.000 |
| damage-visible | material-shift | 1    | 14 | 0.429   | 0.214  | 0.289   | 0.625 | 0.000 |
| damage-visible | material-shift | 2    | 14 | 0.571   | 0.143  | 0.296   | 0.625 | 0.000 |
| damage-visible | material-shift | 3    | 14 | 0.500   | 0.143  | 0.297   | 0.625 | 0.000 |
| damage-visible | material-shift | 4    | 14 | 0.429   | 0.286  | 0.301   | 0.625 | 0.000 |
| damage-visible | material-shift | 5    | 14 | 0.429   | 0.214  | 0.300   | 0.625 | 0.000 |
| damage-visible | material-shift | 6    | 14 | 0.571   | 0.214  | 0.286   | 0.625 | 0.000 |
| damage-visible | material-shift | 7    | 14 | 0.571   | 0.214  | 0.294   | 0.625 | 0.000 |
| ensemble       | material-shift | 0    | 14 | 0.286   | 0.571  | 0.135   | 0.500 | 0.786 |
| ensemble       | material-shift | 1    | 14 | 0.286   | 0.357  | 0.138   | 0.500 | 0.786 |
| ensemble       | material-shift | 2    | 14 | 0.357   | 0.500  | 0.132   | 0.500 | 0.786 |
| ensemble       | material-shift | 3    | 14 | 0.429   | 0.357  | 0.191   | 0.500 | 0.786 |
| ensemble       | material-shift | 4    | 14 | 0.214   | 0.643  | 0.143   | 0.500 | 0.786 |
| ensemble       | material-shift | 5    | 14 | 0.500   | 0.286  | 0.156   | 0.500 | 0.786 |
| ensemble       | material-shift | 6    | 14 | 0.357   | 0.571  | 0.184   | 0.500 | 0.643 |
| ensemble       | material-shift | 7    | 14 | 0.500   | 0.429  | 0.137   | 0.625 | 0.857 |
| graph          | material-shift | 0    | 14 | 0.357   | 0.214  | 0.273   | 0.625 | 0.000 |
| graph          | material-shift | 1    | 14 | 0.214   | 0.214  | 0.311   | 0.625 | 0.000 |

Table 10 continued

| method   | split          | seed | n  | success | damage | mem err | F1    | diag  |
|----------|----------------|------|----|---------|--------|---------|-------|-------|
| graph    | material-shift | 2    | 14 | 0.500   | 0.143  | 0.192   | 0.625 | 0.000 |
| graph    | material-shift | 3    | 14 | 0.571   | 0.071  | 0.280   | 0.625 | 0.000 |
| graph    | material-shift | 4    | 14 | 0.357   | 0.286  | 0.338   | 0.500 | 0.000 |
| graph    | material-shift | 5    | 14 | 0.429   | 0.143  | 0.272   | 0.625 | 0.000 |
| graph    | material-shift | 6    | 14 | 0.500   | 0.214  | 0.240   | 0.625 | 0.000 |
| graph    | material-shift | 7    | 14 | 0.429   | 0.357  | 0.263   | 0.500 | 0.000 |
| history  | material-shift | 0    | 14 | 0.286   | 0.571  | 0.133   | 0.500 | 0.786 |
| history  | material-shift | 1    | 14 | 0.357   | 0.286  | 0.132   | 0.500 | 0.929 |
| history  | material-shift | 2    | 14 | 0.357   | 0.500  | 0.141   | 0.500 | 0.857 |
| history  | material-shift | 3    | 14 | 0.429   | 0.357  | 0.186   | 0.500 | 0.857 |
| history  | material-shift | 4    | 14 | 0.214   | 0.500  | 0.136   | 0.500 | 0.571 |
| history  | material-shift | 5    | 14 | 0.500   | 0.286  | 0.156   | 0.500 | 0.786 |
| history  | material-shift | 6    | 14 | 0.357   | 0.571  | 0.184   | 0.500 | 0.643 |
| history  | material-shift | 7    | 14 | 0.571   | 0.429  | 0.135   | 0.625 | 0.786 |
| oracle   | material-shift | 0    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | material-shift | 1    | 14 | 0.929   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | material-shift | 2    | 14 | 1.000   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | material-shift | 3    | 14 | 1.000   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | material-shift | 4    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.071 |
| oracle   | material-shift | 5    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.071 |
| oracle   | material-shift | 6    | 14 | 0.929   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle   | material-shift | 7    | 14 | 0.929   | 0.000  | 0.000   | 1.000 | 0.000 |
| particle | material-shift | 0    | 14 | 0.214   | 0.643  | 0.133   | 0.625 | 1.000 |
| particle | material-shift | 1    | 14 | 0.357   | 0.357  | 0.132   | 0.500 | 1.000 |
| particle | material-shift | 2    | 14 | 0.357   | 0.500  | 0.141   | 0.500 | 0.857 |
| particle | material-shift | 3    | 14 | 0.500   | 0.357  | 0.186   | 0.625 | 0.857 |
| particle | material-shift | 4    | 14 | 0.214   | 0.643  | 0.136   | 0.625 | 1.000 |
| particle | material-shift | 5    | 14 | 0.429   | 0.429  | 0.156   | 0.500 | 1.000 |
| particle | material-shift | 6    | 14 | 0.214   | 0.714  | 0.184   | 0.625 | 1.000 |
| particle | material-shift | 7    | 14 | 0.500   | 0.429  | 0.135   | 0.625 | 0.929 |

Table 10 continued

| method        | split          | seed | n  | success | damage | mem err | F1    | diag  |
|---------------|----------------|------|----|---------|--------|---------|-------|-------|
| robust-CEM    | material-shift | 0    | 14 | 0.214   | 0.357  | 0.160   | 0.500 | 0.286 |
| robust-CEM    | material-shift | 1    | 14 | 0.286   | 0.286  | 0.163   | 0.500 | 0.357 |
| robust-CEM    | material-shift | 2    | 14 | 0.357   | 0.357  | 0.122   | 0.500 | 0.429 |
| robust-CEM    | material-shift | 3    | 14 | 0.429   | 0.143  | 0.198   | 0.500 | 0.357 |
| robust-CEM    | material-shift | 4    | 14 | 0.286   | 0.357  | 0.179   | 0.500 | 0.214 |
| robust-CEM    | material-shift | 5    | 14 | 0.357   | 0.214  | 0.176   | 0.625 | 0.214 |
| robust-CEM    | material-shift | 6    | 14 | 0.500   | 0.357  | 0.175   | 0.500 | 0.429 |
| robust-CEM    | material-shift | 7    | 14 | 0.500   | 0.357  | 0.145   | 0.500 | 0.500 |
| safe-particle | material-shift | 0    | 14 | 0.286   | 0.286  | 0.160   | 0.625 | 0.000 |
| safe-particle | material-shift | 1    | 14 | 0.286   | 0.286  | 0.161   | 0.500 | 0.143 |
| safe-particle | material-shift | 2    | 14 | 0.500   | 0.143  | 0.122   | 0.625 | 0.000 |
| safe-particle | material-shift | 3    | 14 | 0.429   | 0.143  | 0.197   | 0.625 | 0.071 |
| safe-particle | material-shift | 4    | 14 | 0.286   | 0.357  | 0.180   | 0.500 | 0.071 |
| safe-particle | material-shift | 5    | 14 | 0.286   | 0.143  | 0.177   | 0.625 | 0.000 |
| safe-particle | material-shift | 6    | 14 | 0.429   | 0.286  | 0.175   | 0.500 | 0.143 |
| safe-particle | material-shift | 7    | 14 | 0.500   | 0.286  | 0.146   | 0.500 | 0.000 |
| visible-MPC   | material-shift | 0    | 14 | 0.429   | 0.286  | 0.295   | 0.625 | 0.000 |
| visible-MPC   | material-shift | 1    | 14 | 0.429   | 0.214  | 0.289   | 0.625 | 0.000 |
| visible-MPC   | material-shift | 2    | 14 | 0.571   | 0.143  | 0.296   | 0.625 | 0.000 |
| visible-MPC   | material-shift | 3    | 14 | 0.500   | 0.143  | 0.297   | 0.625 | 0.000 |
| visible-MPC   | material-shift | 4    | 14 | 0.429   | 0.286  | 0.301   | 0.625 | 0.000 |
| visible-MPC   | material-shift | 5    | 14 | 0.429   | 0.214  | 0.300   | 0.625 | 0.000 |
| visible-MPC   | material-shift | 6    | 14 | 0.571   | 0.214  | 0.286   | 0.625 | 0.000 |
| visible-MPC   | material-shift | 7    | 14 | 0.571   | 0.214  | 0.294   | 0.625 | 0.000 |
| ACM-v4        | nominal        | 0    | 14 | 0.214   | 0.286  | 0.103   | 0.625 | 0.214 |
| ACM-v4        | nominal        | 1    | 14 | 0.357   | 0.214  | 0.082   | 0.625 | 0.143 |
| ACM-v4        | nominal        | 2    | 14 | 0.357   | 0.214  | 0.118   | 0.625 | 0.214 |
| ACM-v4        | nominal        | 3    | 14 | 0.429   | 0.286  | 0.134   | 0.625 | 0.143 |
| ACM-v4        | nominal        | 4    | 14 | 0.286   | 0.286  | 0.165   | 0.625 | 0.286 |
| ACM-v4        | nominal        | 5    | 14 | 0.286   | 0.214  | 0.079   | 0.625 | 0.071 |

Table 10 continued

| method         | split   | seed | n  | success | damage | mem err | F1    | diag  |
|----------------|---------|------|----|---------|--------|---------|-------|-------|
| ACM-v4         | nominal | 6    | 14 | 0.571   | 0.286  | 0.073   | 0.500 | 0.286 |
| ACM-v4         | nominal | 7    | 14 | 0.429   | 0.286  | 0.131   | 0.625 | 0.357 |
| ACM-v5         | nominal | 0    | 14 | 0.286   | 0.286  | 0.095   | 0.625 | 0.000 |
| ACM-v5         | nominal | 1    | 14 | 0.286   | 0.214  | 0.074   | 0.625 | 0.000 |
| ACM-v5         | nominal | 2    | 14 | 0.357   | 0.214  | 0.110   | 0.625 | 0.000 |
| ACM-v5         | nominal | 3    | 14 | 0.429   | 0.286  | 0.129   | 0.625 | 0.000 |
| ACM-v5         | nominal | 4    | 14 | 0.286   | 0.214  | 0.153   | 0.625 | 0.000 |
| ACM-v5         | nominal | 5    | 14 | 0.286   | 0.214  | 0.076   | 0.625 | 0.000 |
| ACM-v5         | nominal | 6    | 14 | 0.500   | 0.143  | 0.067   | 0.625 | 0.000 |
| ACM-v5         | nominal | 7    | 14 | 0.500   | 0.286  | 0.124   | 0.625 | 0.000 |
| damage-visible | nominal | 0    | 14 | 0.286   | 0.286  | 0.119   | 0.625 | 0.000 |
| damage-visible | nominal | 1    | 14 | 0.357   | 0.214  | 0.122   | 0.625 | 0.000 |
| damage-visible | nominal | 2    | 14 | 0.357   | 0.214  | 0.126   | 0.625 | 0.000 |
| damage-visible | nominal | 3    | 14 | 0.429   | 0.286  | 0.129   | 0.625 | 0.000 |
| damage-visible | nominal | 4    | 14 | 0.357   | 0.286  | 0.127   | 0.625 | 0.000 |
| damage-visible | nominal | 5    | 14 | 0.286   | 0.214  | 0.119   | 0.625 | 0.000 |
| damage-visible | nominal | 6    | 14 | 0.571   | 0.214  | 0.118   | 0.625 | 0.000 |
| damage-visible | nominal | 7    | 14 | 0.500   | 0.286  | 0.129   | 0.625 | 0.000 |
| ensemble       | nominal | 0    | 14 | 0.214   | 0.286  | 0.132   | 0.625 | 0.357 |
| ensemble       | nominal | 1    | 14 | 0.357   | 0.214  | 0.111   | 0.625 | 0.357 |
| ensemble       | nominal | 2    | 14 | 0.286   | 0.286  | 0.124   | 0.500 | 0.429 |
| ensemble       | nominal | 3    | 14 | 0.429   | 0.286  | 0.144   | 0.625 | 0.286 |
| ensemble       | nominal | 4    | 14 | 0.286   | 0.286  | 0.189   | 0.625 | 0.357 |
| ensemble       | nominal | 5    | 14 | 0.357   | 0.214  | 0.088   | 0.625 | 0.143 |
| ensemble       | nominal | 6    | 14 | 0.429   | 0.286  | 0.084   | 0.500 | 0.429 |
| ensemble       | nominal | 7    | 14 | 0.429   | 0.286  | 0.149   | 0.625 | 0.357 |
| graph          | nominal | 0    | 14 | 0.286   | 0.286  | 0.138   | 0.625 | 0.000 |
| graph          | nominal | 1    | 14 | 0.357   | 0.214  | 0.096   | 0.625 | 0.000 |
| graph          | nominal | 2    | 14 | 0.357   | 0.214  | 0.114   | 0.625 | 0.000 |
| graph          | nominal | 3    | 14 | 0.500   | 0.214  | 0.116   | 0.625 | 0.000 |

Table 10 continued

| method     | split   | seed | n  | success | damage | mem err | F1    | diag  |
|------------|---------|------|----|---------|--------|---------|-------|-------|
| graph      | nominal | 4    | 14 | 0.286   | 0.286  | 0.088   | 0.625 | 0.000 |
| graph      | nominal | 5    | 14 | 0.286   | 0.214  | 0.096   | 0.625 | 0.000 |
| graph      | nominal | 6    | 14 | 0.500   | 0.071  | 0.128   | 0.625 | 0.000 |
| graph      | nominal | 7    | 14 | 0.500   | 0.214  | 0.093   | 0.625 | 0.000 |
| history    | nominal | 0    | 14 | 0.214   | 0.286  | 0.127   | 0.625 | 0.214 |
| history    | nominal | 1    | 14 | 0.357   | 0.214  | 0.100   | 0.625 | 0.214 |
| history    | nominal | 2    | 14 | 0.357   | 0.214  | 0.134   | 0.625 | 0.286 |
| history    | nominal | 3    | 14 | 0.429   | 0.286  | 0.148   | 0.625 | 0.286 |
| history    | nominal | 4    | 14 | 0.286   | 0.286  | 0.174   | 0.625 | 0.286 |
| history    | nominal | 5    | 14 | 0.286   | 0.214  | 0.095   | 0.625 | 0.071 |
| history    | nominal | 6    | 14 | 0.429   | 0.286  | 0.084   | 0.500 | 0.286 |
| history    | nominal | 7    | 14 | 0.429   | 0.286  | 0.143   | 0.625 | 0.357 |
| oracle     | nominal | 0    | 14 | 0.643   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | nominal | 1    | 14 | 0.643   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | nominal | 2    | 14 | 0.643   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | nominal | 3    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | nominal | 4    | 14 | 0.643   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | nominal | 5    | 14 | 0.643   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | nominal | 6    | 14 | 0.786   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | nominal | 7    | 14 | 0.786   | 0.000  | 0.000   | 1.000 | 0.000 |
| particle   | nominal | 0    | 14 | 0.286   | 0.357  | 0.127   | 0.500 | 0.357 |
| particle   | nominal | 1    | 14 | 0.286   | 0.286  | 0.100   | 0.500 | 0.643 |
| particle   | nominal | 2    | 14 | 0.286   | 0.286  | 0.134   | 0.500 | 0.643 |
| particle   | nominal | 3    | 14 | 0.429   | 0.357  | 0.148   | 0.500 | 0.643 |
| particle   | nominal | 4    | 14 | 0.286   | 0.286  | 0.174   | 0.625 | 0.500 |
| particle   | nominal | 5    | 14 | 0.357   | 0.214  | 0.095   | 0.625 | 0.429 |
| particle   | nominal | 6    | 14 | 0.429   | 0.286  | 0.084   | 0.500 | 0.714 |
| particle   | nominal | 7    | 14 | 0.429   | 0.286  | 0.143   | 0.625 | 0.429 |
| robust-CEM | nominal | 0    | 14 | 0.214   | 0.286  | 0.093   | 0.625 | 0.071 |
| robust-CEM | nominal | 1    | 14 | 0.286   | 0.214  | 0.070   | 0.625 | 0.000 |

Table 10 continued

| method        | split            | seed | n  | success | damage | mem err | F1    | diag  |
|---------------|------------------|------|----|---------|--------|---------|-------|-------|
| robust-CEM    | nominal          | 2    | 14 | 0.357   | 0.214  | 0.106   | 0.625 | 0.071 |
| robust-CEM    | nominal          | 3    | 14 | 0.429   | 0.286  | 0.124   | 0.625 | 0.071 |
| robust-CEM    | nominal          | 4    | 14 | 0.286   | 0.286  | 0.142   | 0.625 | 0.071 |
| robust-CEM    | nominal          | 5    | 14 | 0.286   | 0.214  | 0.074   | 0.625 | 0.000 |
| robust-CEM    | nominal          | 6    | 14 | 0.500   | 0.214  | 0.067   | 0.625 | 0.000 |
| robust-CEM    | nominal          | 7    | 14 | 0.500   | 0.286  | 0.117   | 0.625 | 0.071 |
| safe-particle | nominal          | 0    | 14 | 0.286   | 0.286  | 0.095   | 0.625 | 0.000 |
| safe-particle | nominal          | 1    | 14 | 0.286   | 0.214  | 0.069   | 0.625 | 0.000 |
| safe-particle | nominal          | 2    | 14 | 0.357   | 0.214  | 0.105   | 0.625 | 0.000 |
| safe-particle | nominal          | 3    | 14 | 0.429   | 0.286  | 0.125   | 0.625 | 0.000 |
| safe-particle | nominal          | 4    | 14 | 0.286   | 0.286  | 0.136   | 0.625 | 0.000 |
| safe-particle | nominal          | 5    | 14 | 0.286   | 0.214  | 0.072   | 0.625 | 0.000 |
| safe-particle | nominal          | 6    | 14 | 0.500   | 0.214  | 0.068   | 0.625 | 0.000 |
| safe-particle | nominal          | 7    | 14 | 0.500   | 0.286  | 0.117   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 0    | 14 | 0.286   | 0.286  | 0.119   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 1    | 14 | 0.357   | 0.214  | 0.122   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 2    | 14 | 0.357   | 0.214  | 0.126   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 3    | 14 | 0.429   | 0.286  | 0.129   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 4    | 14 | 0.357   | 0.286  | 0.127   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 5    | 14 | 0.286   | 0.214  | 0.119   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 6    | 14 | 0.571   | 0.214  | 0.118   | 0.625 | 0.000 |
| visible-MPC   | nominal          | 7    | 14 | 0.500   | 0.286  | 0.129   | 0.625 | 0.000 |
| ACM-v4        | occluded-contact | 0    | 14 | 0.500   | 0.357  | 0.222   | 0.500 | 0.786 |
| ACM-v4        | occluded-contact | 1    | 14 | 0.214   | 0.500  | 0.205   | 0.375 | 0.857 |
| ACM-v4        | occluded-contact | 2    | 14 | 0.429   | 0.500  | 0.163   | 0.500 | 0.857 |
| ACM-v4        | occluded-contact | 3    | 14 | 0.429   | 0.500  | 0.167   | 0.375 | 0.857 |
| ACM-v4        | occluded-contact | 4    | 14 | 0.357   | 0.500  | 0.171   | 0.375 | 0.857 |
| ACM-v4        | occluded-contact | 5    | 14 | 0.429   | 0.429  | 0.254   | 0.375 | 0.643 |
| ACM-v4        | occluded-contact | 6    | 14 | 0.429   | 0.500  | 0.218   | 0.500 | 0.786 |
| ACM-v4        | occluded-contact | 7    | 14 | 0.286   | 0.571  | 0.192   | 0.375 | 0.714 |

Table 10 continued

| method         | split            | seed | n  | success | damage | mem err | F1    | diag  |
|----------------|------------------|------|----|---------|--------|---------|-------|-------|
| ACM-v5         | occluded-contact | 0    | 14 | 0.357   | 0.286  | 0.233   | 0.500 | 0.000 |
| ACM-v5         | occluded-contact | 1    | 14 | 0.429   | 0.214  | 0.202   | 0.500 | 0.000 |
| ACM-v5         | occluded-contact | 2    | 14 | 0.429   | 0.286  | 0.158   | 0.500 | 0.000 |
| ACM-v5         | occluded-contact | 3    | 14 | 0.500   | 0.214  | 0.179   | 0.375 | 0.000 |
| ACM-v5         | occluded-contact | 4    | 14 | 0.429   | 0.286  | 0.178   | 0.500 | 0.000 |
| ACM-v5         | occluded-contact | 5    | 14 | 0.286   | 0.357  | 0.261   | 0.375 | 0.000 |
| ACM-v5         | occluded-contact | 6    | 14 | 0.571   | 0.357  | 0.221   | 0.375 | 0.000 |
| ACM-v5         | occluded-contact | 7    | 14 | 0.357   | 0.357  | 0.190   | 0.375 | 0.000 |
| damage-visible | occluded-contact | 0    | 14 | 0.500   | 0.214  | 0.370   | 0.375 | 0.000 |
| damage-visible | occluded-contact | 1    | 14 | 0.571   | 0.214  | 0.410   | 0.500 | 0.000 |
| damage-visible | occluded-contact | 2    | 14 | 0.643   | 0.214  | 0.407   | 0.500 | 0.000 |
| damage-visible | occluded-contact | 3    | 14 | 0.571   | 0.286  | 0.401   | 0.375 | 0.000 |
| damage-visible | occluded-contact | 4    | 14 | 0.500   | 0.286  | 0.386   | 0.500 | 0.000 |
| damage-visible | occluded-contact | 5    | 14 | 0.429   | 0.286  | 0.388   | 0.375 | 0.000 |
| damage-visible | occluded-contact | 6    | 14 | 0.571   | 0.214  | 0.394   | 0.500 | 0.000 |
| damage-visible | occluded-contact | 7    | 14 | 0.500   | 0.286  | 0.388   | 0.500 | 0.000 |
| ensemble       | occluded-contact | 0    | 14 | 0.500   | 0.357  | 0.213   | 0.500 | 0.714 |
| ensemble       | occluded-contact | 1    | 14 | 0.214   | 0.500  | 0.202   | 0.375 | 0.857 |
| ensemble       | occluded-contact | 2    | 14 | 0.429   | 0.500  | 0.186   | 0.500 | 0.857 |
| ensemble       | occluded-contact | 3    | 14 | 0.429   | 0.571  | 0.164   | 0.500 | 0.929 |
| ensemble       | occluded-contact | 4    | 14 | 0.357   | 0.500  | 0.163   | 0.375 | 0.857 |
| ensemble       | occluded-contact | 5    | 14 | 0.429   | 0.500  | 0.275   | 0.375 | 0.714 |
| ensemble       | occluded-contact | 6    | 14 | 0.429   | 0.500  | 0.210   | 0.500 | 0.857 |
| ensemble       | occluded-contact | 7    | 14 | 0.286   | 0.500  | 0.189   | 0.375 | 0.714 |
| graph          | occluded-contact | 0    | 14 | 0.357   | 0.286  | 0.384   | 0.375 | 0.000 |
| graph          | occluded-contact | 1    | 14 | 0.500   | 0.214  | 0.403   | 0.500 | 0.000 |
| graph          | occluded-contact | 2    | 14 | 0.500   | 0.357  | 0.248   | 0.375 | 0.000 |
| graph          | occluded-contact | 3    | 14 | 0.357   | 0.286  | 0.366   | 0.375 | 0.000 |
| graph          | occluded-contact | 4    | 14 | 0.429   | 0.357  | 0.336   | 0.375 | 0.000 |
| graph          | occluded-contact | 5    | 14 | 0.357   | 0.357  | 0.391   | 0.375 | 0.000 |

Table 10 continued

| method     | split            | seed | n  | success | damage | mem err | F1    | diag  |
|------------|------------------|------|----|---------|--------|---------|-------|-------|
| graph      | occluded-contact | 6    | 14 | 0.429   | 0.286  | 0.411   | 0.375 | 0.000 |
| graph      | occluded-contact | 7    | 14 | 0.429   | 0.286  | 0.313   | 0.375 | 0.000 |
| history    | occluded-contact | 0    | 14 | 0.500   | 0.357  | 0.203   | 0.500 | 0.786 |
| history    | occluded-contact | 1    | 14 | 0.286   | 0.429  | 0.208   | 0.375 | 0.786 |
| history    | occluded-contact | 2    | 14 | 0.429   | 0.500  | 0.177   | 0.500 | 0.929 |
| history    | occluded-contact | 3    | 14 | 0.429   | 0.571  | 0.153   | 0.500 | 1.000 |
| history    | occluded-contact | 4    | 14 | 0.357   | 0.500  | 0.166   | 0.375 | 0.857 |
| history    | occluded-contact | 5    | 14 | 0.357   | 0.643  | 0.264   | 0.500 | 0.786 |
| history    | occluded-contact | 6    | 14 | 0.429   | 0.500  | 0.210   | 0.500 | 0.857 |
| history    | occluded-contact | 7    | 14 | 0.357   | 0.500  | 0.193   | 0.375 | 0.714 |
| oracle     | occluded-contact | 0    | 14 | 0.929   | 0.071  | 0.000   | 1.000 | 0.000 |
| oracle     | occluded-contact | 1    | 14 | 1.000   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | occluded-contact | 2    | 14 | 0.714   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | occluded-contact | 3    | 14 | 0.857   | 0.071  | 0.000   | 1.000 | 0.000 |
| oracle     | occluded-contact | 4    | 14 | 0.786   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | occluded-contact | 5    | 14 | 0.857   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | occluded-contact | 6    | 14 | 0.786   | 0.000  | 0.000   | 1.000 | 0.000 |
| oracle     | occluded-contact | 7    | 14 | 0.714   | 0.000  | 0.000   | 1.000 | 0.000 |
| particle   | occluded-contact | 0    | 14 | 0.500   | 0.357  | 0.203   | 0.500 | 0.929 |
| particle   | occluded-contact | 1    | 14 | 0.214   | 0.571  | 0.208   | 0.500 | 0.929 |
| particle   | occluded-contact | 2    | 14 | 0.429   | 0.500  | 0.177   | 0.500 | 0.929 |
| particle   | occluded-contact | 3    | 14 | 0.429   | 0.571  | 0.153   | 0.500 | 0.929 |
| particle   | occluded-contact | 4    | 14 | 0.357   | 0.571  | 0.166   | 0.500 | 1.000 |
| particle   | occluded-contact | 5    | 14 | 0.286   | 0.643  | 0.264   | 0.375 | 0.929 |
| particle   | occluded-contact | 6    | 14 | 0.429   | 0.500  | 0.210   | 0.500 | 0.857 |
| particle   | occluded-contact | 7    | 14 | 0.214   | 0.643  | 0.193   | 0.375 | 0.929 |
| robust-CEM | occluded-contact | 0    | 14 | 0.500   | 0.286  | 0.247   | 0.375 | 0.357 |
| robust-CEM | occluded-contact | 1    | 14 | 0.357   | 0.214  | 0.200   | 0.500 | 0.357 |
| robust-CEM | occluded-contact | 2    | 14 | 0.429   | 0.357  | 0.156   | 0.500 | 0.500 |
| robust-CEM | occluded-contact | 3    | 14 | 0.500   | 0.357  | 0.191   | 0.375 | 0.500 |

Table 10 continued

| method        | split            | seed | n  | success | damage | mem err | F1    | diag  |
|---------------|------------------|------|----|---------|--------|---------|-------|-------|
| robust-CEM    | occluded-contact | 4    | 14 | 0.357   | 0.357  | 0.192   | 0.375 | 0.500 |
| robust-CEM    | occluded-contact | 5    | 14 | 0.357   | 0.429  | 0.274   | 0.375 | 0.214 |
| robust-CEM    | occluded-contact | 6    | 14 | 0.500   | 0.286  | 0.233   | 0.375 | 0.214 |
| robust-CEM    | occluded-contact | 7    | 14 | 0.357   | 0.429  | 0.197   | 0.375 | 0.429 |
| safe-particle | occluded-contact | 0    | 14 | 0.429   | 0.286  | 0.248   | 0.375 | 0.000 |
| safe-particle | occluded-contact | 1    | 14 | 0.429   | 0.286  | 0.198   | 0.375 | 0.071 |
| safe-particle | occluded-contact | 2    | 14 | 0.429   | 0.357  | 0.157   | 0.375 | 0.214 |
| safe-particle | occluded-contact | 3    | 14 | 0.500   | 0.286  | 0.191   | 0.375 | 0.071 |
| safe-particle | occluded-contact | 4    | 14 | 0.429   | 0.286  | 0.194   | 0.500 | 0.143 |
| safe-particle | occluded-contact | 5    | 14 | 0.357   | 0.357  | 0.274   | 0.375 | 0.000 |
| safe-particle | occluded-contact | 6    | 14 | 0.500   | 0.286  | 0.231   | 0.375 | 0.000 |
| safe-particle | occluded-contact | 7    | 14 | 0.500   | 0.214  | 0.196   | 0.500 | 0.071 |
| visible-MPC   | occluded-contact | 0    | 14 | 0.500   | 0.214  | 0.370   | 0.375 | 0.000 |
| visible-MPC   | occluded-contact | 1    | 14 | 0.571   | 0.214  | 0.410   | 0.500 | 0.000 |
| visible-MPC   | occluded-contact | 2    | 14 | 0.643   | 0.214  | 0.407   | 0.500 | 0.000 |
| visible-MPC   | occluded-contact | 3    | 14 | 0.571   | 0.286  | 0.401   | 0.375 | 0.000 |
| visible-MPC   | occluded-contact | 4    | 14 | 0.500   | 0.286  | 0.386   | 0.500 | 0.000 |
| visible-MPC   | occluded-contact | 5    | 14 | 0.429   | 0.286  | 0.388   | 0.375 | 0.000 |
| visible-MPC   | occluded-contact | 6    | 14 | 0.571   | 0.214  | 0.394   | 0.500 | 0.000 |
| visible-MPC   | occluded-contact | 7    | 14 | 0.500   | 0.286  | 0.388   | 0.500 | 0.000 |

## E PER-SEED FIXED-RISK METRICS

The fixed-risk gate is decisive for this paper because v5 improves latent-memory proxies but can still fail safety. Table 11 includes seed-level fixed-risk evidence for the two frozen promotion budgets, 0.10 and 0.20.

Table 11: Per-seed fixed-risk metrics at the two decisive budgets.

| method         | split    | budget | seed | n | success | damage | clip  | diag  |
|----------------|----------|--------|------|---|---------|--------|-------|-------|
| ACM-v4         | combined | 0.10   | 0    | 8 | 0.375   | 0.625  | 0.027 | 0.250 |
| ACM-v4         | combined | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| ACM-v4         | combined | 0.10   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.125 |
| ACM-v4         | combined | 0.10   | 3    | 8 | 0.625   | 0.000  | 0.021 | 0.000 |
| ACM-v4         | combined | 0.10   | 4    | 8 | 0.750   | 0.250  | 0.025 | 0.125 |
| ACM-v4         | combined | 0.10   | 5    | 8 | 0.500   | 0.125  | 0.021 | 0.000 |
| ACM-v4         | combined | 0.10   | 6    | 8 | 0.500   | 0.125  | 0.023 | 0.000 |
| ACM-v4         | combined | 0.10   | 7    | 8 | 0.500   | 0.250  | 0.027 | 0.125 |
| ACM-v5         | combined | 0.10   | 0    | 8 | 0.750   | 0.250  | 0.031 | 0.000 |
| ACM-v5         | combined | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| ACM-v5         | combined | 0.10   | 2    | 8 | 0.500   | 0.375  | 0.040 | 0.000 |
| ACM-v5         | combined | 0.10   | 3    | 8 | 0.625   | 0.000  | 0.025 | 0.000 |
| ACM-v5         | combined | 0.10   | 4    | 8 | 0.875   | 0.125  | 0.029 | 0.000 |
| ACM-v5         | combined | 0.10   | 5    | 8 | 0.625   | 0.000  | 0.033 | 0.000 |
| ACM-v5         | combined | 0.10   | 6    | 8 | 0.500   | 0.125  | 0.027 | 0.000 |
| ACM-v5         | combined | 0.10   | 7    | 8 | 0.500   | 0.250  | 0.031 | 0.000 |
| damage-visible | combined | 0.10   | 0    | 8 | 0.875   | 0.125  | 0.044 | 0.000 |
| damage-visible | combined | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| damage-visible | combined | 0.10   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.000 |
| damage-visible | combined | 0.10   | 3    | 8 | 0.625   | 0.125  | 0.037 | 0.000 |
| damage-visible | combined | 0.10   | 4    | 8 | 0.875   | 0.125  | 0.035 | 0.000 |
| damage-visible | combined | 0.10   | 5    | 8 | 0.625   | 0.000  | 0.040 | 0.000 |
| damage-visible | combined | 0.10   | 6    | 8 | 0.375   | 0.250  | 0.035 | 0.000 |
| damage-visible | combined | 0.10   | 7    | 8 | 0.500   | 0.250  | 0.035 | 0.000 |
| ensemble       | combined | 0.10   | 0    | 8 | 0.375   | 0.625  | 0.027 | 0.250 |
| ensemble       | combined | 0.10   | 1    | 8 | 0.500   | 0.125  | 0.015 | 0.000 |
| ensemble       | combined | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.035 | 0.125 |
| ensemble       | combined | 0.10   | 3    | 8 | 0.500   | 0.125  | 0.015 | 0.000 |
| ensemble       | combined | 0.10   | 4    | 8 | 0.875   | 0.125  | 0.027 | 0.125 |
| ensemble       | combined | 0.10   | 5    | 8 | 0.500   | 0.125  | 0.017 | 0.000 |

Table 11 continued

| method        | split    | budget | seed | n | success | damage | clip  | diag  |
|---------------|----------|--------|------|---|---------|--------|-------|-------|
| ensemble      | combined | 0.10   | 6    | 8 | 0.625   | 0.250  | 0.023 | 0.000 |
| ensemble      | combined | 0.10   | 7    | 8 | 0.375   | 0.250  | 0.019 | 0.125 |
| oracle        | combined | 0.10   | 0    | 8 | 0.875   | 0.125  | 0.040 | 0.000 |
| oracle        | combined | 0.10   | 1    | 8 | 0.875   | 0.000  | 0.015 | 0.000 |
| oracle        | combined | 0.10   | 2    | 8 | 0.750   | 0.000  | 0.017 | 0.000 |
| oracle        | combined | 0.10   | 3    | 8 | 0.875   | 0.000  | 0.023 | 0.000 |
| oracle        | combined | 0.10   | 4    | 8 | 0.875   | 0.125  | 0.031 | 0.000 |
| oracle        | combined | 0.10   | 5    | 8 | 0.750   | 0.000  | 0.029 | 0.000 |
| oracle        | combined | 0.10   | 6    | 8 | 0.875   | 0.000  | 0.021 | 0.000 |
| oracle        | combined | 0.10   | 7    | 8 | 0.750   | 0.125  | 0.019 | 0.000 |
| particle      | combined | 0.10   | 0    | 8 | 0.375   | 0.625  | 0.037 | 0.500 |
| particle      | combined | 0.10   | 1    | 8 | 0.375   | 0.375  | 0.040 | 0.250 |
| particle      | combined | 0.10   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.125 |
| particle      | combined | 0.10   | 3    | 8 | 0.375   | 0.250  | 0.027 | 0.250 |
| particle      | combined | 0.10   | 4    | 8 | 0.750   | 0.250  | 0.035 | 0.250 |
| particle      | combined | 0.10   | 5    | 8 | 0.375   | 0.250  | 0.033 | 0.250 |
| particle      | combined | 0.10   | 6    | 8 | 0.375   | 0.375  | 0.037 | 0.375 |
| particle      | combined | 0.10   | 7    | 8 | 0.375   | 0.375  | 0.031 | 0.375 |
| robust-CEM    | combined | 0.10   | 0    | 8 | 0.500   | 0.500  | 0.033 | 0.250 |
| robust-CEM    | combined | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| robust-CEM    | combined | 0.10   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.125 |
| robust-CEM    | combined | 0.10   | 3    | 8 | 0.625   | 0.125  | 0.031 | 0.000 |
| robust-CEM    | combined | 0.10   | 4    | 8 | 0.875   | 0.125  | 0.031 | 0.125 |
| robust-CEM    | combined | 0.10   | 5    | 8 | 0.625   | 0.000  | 0.033 | 0.000 |
| robust-CEM    | combined | 0.10   | 6    | 8 | 0.500   | 0.125  | 0.027 | 0.000 |
| robust-CEM    | combined | 0.10   | 7    | 8 | 0.500   | 0.250  | 0.033 | 0.125 |
| safe-particle | combined | 0.10   | 0    | 8 | 0.375   | 0.625  | 0.031 | 0.375 |
| safe-particle | combined | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| safe-particle | combined | 0.10   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.125 |
| safe-particle | combined | 0.10   | 3    | 8 | 0.500   | 0.125  | 0.027 | 0.125 |

Table 11 continued

| method         | split    | budget | seed | n | success | damage | clip  | diag  |
|----------------|----------|--------|------|---|---------|--------|-------|-------|
| safe-particle  | combined | 0.10   | 4    | 8 | 0.750   | 0.250  | 0.035 | 0.250 |
| safe-particle  | combined | 0.10   | 5    | 8 | 0.625   | 0.000  | 0.029 | 0.000 |
| safe-particle  | combined | 0.10   | 6    | 8 | 0.500   | 0.125  | 0.033 | 0.000 |
| safe-particle  | combined | 0.10   | 7    | 8 | 0.500   | 0.250  | 0.033 | 0.125 |
| visible-MPC    | combined | 0.10   | 0    | 8 | 0.875   | 0.125  | 0.044 | 0.000 |
| visible-MPC    | combined | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| visible-MPC    | combined | 0.10   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.000 |
| visible-MPC    | combined | 0.10   | 3    | 8 | 0.625   | 0.125  | 0.037 | 0.000 |
| visible-MPC    | combined | 0.10   | 4    | 8 | 0.875   | 0.125  | 0.035 | 0.000 |
| visible-MPC    | combined | 0.10   | 5    | 8 | 0.625   | 0.000  | 0.040 | 0.000 |
| visible-MPC    | combined | 0.10   | 6    | 8 | 0.375   | 0.250  | 0.035 | 0.000 |
| visible-MPC    | combined | 0.10   | 7    | 8 | 0.500   | 0.250  | 0.035 | 0.000 |
| ACM-v4         | combined | 0.20   | 0    | 8 | 0.125   | 0.875  | 0.035 | 1.000 |
| ACM-v4         | combined | 0.20   | 1    | 8 | 0.375   | 0.500  | 0.042 | 0.875 |
| ACM-v4         | combined | 0.20   | 2    | 8 | 0.625   | 0.375  | 0.040 | 1.000 |
| ACM-v4         | combined | 0.20   | 3    | 8 | 0.125   | 0.750  | 0.033 | 1.000 |
| ACM-v4         | combined | 0.20   | 4    | 8 | 0.375   | 0.500  | 0.035 | 1.000 |
| ACM-v4         | combined | 0.20   | 5    | 8 | 0.125   | 0.875  | 0.033 | 1.000 |
| ACM-v4         | combined | 0.20   | 6    | 8 | 0.375   | 0.500  | 0.035 | 1.000 |
| ACM-v4         | combined | 0.20   | 7    | 8 | 0.375   | 0.625  | 0.029 | 1.000 |
| ACM-v5         | combined | 0.20   | 0    | 8 | 0.750   | 0.250  | 0.031 | 0.000 |
| ACM-v5         | combined | 0.20   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| ACM-v5         | combined | 0.20   | 2    | 8 | 0.500   | 0.375  | 0.040 | 0.000 |
| ACM-v5         | combined | 0.20   | 3    | 8 | 0.625   | 0.000  | 0.025 | 0.000 |
| ACM-v5         | combined | 0.20   | 4    | 8 | 0.875   | 0.125  | 0.029 | 0.000 |
| ACM-v5         | combined | 0.20   | 5    | 8 | 0.625   | 0.000  | 0.033 | 0.000 |
| ACM-v5         | combined | 0.20   | 6    | 8 | 0.500   | 0.125  | 0.027 | 0.000 |
| ACM-v5         | combined | 0.20   | 7    | 8 | 0.500   | 0.250  | 0.031 | 0.000 |
| damage-visible | combined | 0.20   | 0    | 8 | 0.875   | 0.125  | 0.044 | 0.000 |
| damage-visible | combined | 0.20   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |

Table 11 continued

| method         | split    | budget | seed | n | success | damage | clip  | diag  |
|----------------|----------|--------|------|---|---------|--------|-------|-------|
| damage-visible | combined | 0.20   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.000 |
| damage-visible | combined | 0.20   | 3    | 8 | 0.625   | 0.125  | 0.037 | 0.000 |
| damage-visible | combined | 0.20   | 4    | 8 | 0.875   | 0.125  | 0.035 | 0.000 |
| damage-visible | combined | 0.20   | 5    | 8 | 0.625   | 0.000  | 0.040 | 0.000 |
| damage-visible | combined | 0.20   | 6    | 8 | 0.375   | 0.250  | 0.035 | 0.000 |
| damage-visible | combined | 0.20   | 7    | 8 | 0.500   | 0.250  | 0.035 | 0.000 |
| ensemble       | combined | 0.20   | 0    | 8 | 0.125   | 0.875  | 0.035 | 1.000 |
| ensemble       | combined | 0.20   | 1    | 8 | 0.375   | 0.500  | 0.044 | 1.000 |
| ensemble       | combined | 0.20   | 2    | 8 | 0.625   | 0.375  | 0.040 | 1.000 |
| ensemble       | combined | 0.20   | 3    | 8 | 0.250   | 0.625  | 0.035 | 0.875 |
| ensemble       | combined | 0.20   | 4    | 8 | 0.375   | 0.500  | 0.035 | 1.000 |
| ensemble       | combined | 0.20   | 5    | 8 | 0.125   | 0.875  | 0.033 | 1.000 |
| ensemble       | combined | 0.20   | 6    | 8 | 0.375   | 0.500  | 0.035 | 1.000 |
| ensemble       | combined | 0.20   | 7    | 8 | 0.375   | 0.625  | 0.029 | 1.000 |
| oracle         | combined | 0.20   | 0    | 8 | 0.875   | 0.125  | 0.040 | 0.000 |
| oracle         | combined | 0.20   | 1    | 8 | 0.875   | 0.000  | 0.015 | 0.000 |
| oracle         | combined | 0.20   | 2    | 8 | 0.750   | 0.000  | 0.017 | 0.000 |
| oracle         | combined | 0.20   | 3    | 8 | 0.875   | 0.000  | 0.027 | 0.000 |
| oracle         | combined | 0.20   | 4    | 8 | 0.875   | 0.125  | 0.031 | 0.000 |
| oracle         | combined | 0.20   | 5    | 8 | 0.750   | 0.000  | 0.031 | 0.000 |
| oracle         | combined | 0.20   | 6    | 8 | 0.875   | 0.000  | 0.021 | 0.000 |
| oracle         | combined | 0.20   | 7    | 8 | 0.750   | 0.125  | 0.019 | 0.000 |
| particle       | combined | 0.20   | 0    | 8 | 0.125   | 0.875  | 0.035 | 1.000 |
| particle       | combined | 0.20   | 1    | 8 | 0.375   | 0.500  | 0.042 | 0.875 |
| particle       | combined | 0.20   | 2    | 8 | 0.625   | 0.375  | 0.040 | 1.000 |
| particle       | combined | 0.20   | 3    | 8 | 0.125   | 0.750  | 0.033 | 1.000 |
| particle       | combined | 0.20   | 4    | 8 | 0.375   | 0.500  | 0.035 | 1.000 |
| particle       | combined | 0.20   | 5    | 8 | 0.125   | 0.875  | 0.033 | 1.000 |
| particle       | combined | 0.20   | 6    | 8 | 0.375   | 0.500  | 0.035 | 1.000 |
| particle       | combined | 0.20   | 7    | 8 | 0.375   | 0.625  | 0.029 | 1.000 |

Table 11 continued

| method        | split         | budget | seed | n | success | damage | clip  | diag  |
|---------------|---------------|--------|------|---|---------|--------|-------|-------|
| robust-CEM    | combined      | 0.20   | 0    | 8 | 0.375   | 0.625  | 0.035 | 0.750 |
| robust-CEM    | combined      | 0.20   | 1    | 8 | 0.500   | 0.250  | 0.042 | 0.625 |
| robust-CEM    | combined      | 0.20   | 2    | 8 | 0.625   | 0.375  | 0.040 | 1.000 |
| robust-CEM    | combined      | 0.20   | 3    | 8 | 0.375   | 0.500  | 0.040 | 0.750 |
| robust-CEM    | combined      | 0.20   | 4    | 8 | 0.500   | 0.375  | 0.035 | 0.750 |
| robust-CEM    | combined      | 0.20   | 5    | 8 | 0.250   | 0.750  | 0.037 | 0.875 |
| robust-CEM    | combined      | 0.20   | 6    | 8 | 0.375   | 0.375  | 0.033 | 0.750 |
| robust-CEM    | combined      | 0.20   | 7    | 8 | 0.500   | 0.375  | 0.035 | 0.625 |
| safe-particle | combined      | 0.20   | 0    | 8 | 0.375   | 0.625  | 0.031 | 0.500 |
| safe-particle | combined      | 0.20   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.250 |
| safe-particle | combined      | 0.20   | 2    | 8 | 0.625   | 0.375  | 0.040 | 0.625 |
| safe-particle | combined      | 0.20   | 3    | 8 | 0.500   | 0.250  | 0.027 | 0.375 |
| safe-particle | combined      | 0.20   | 4    | 8 | 0.625   | 0.375  | 0.035 | 0.625 |
| safe-particle | combined      | 0.20   | 5    | 8 | 0.500   | 0.250  | 0.029 | 0.250 |
| safe-particle | combined      | 0.20   | 6    | 8 | 0.500   | 0.125  | 0.029 | 0.375 |
| safe-particle | combined      | 0.20   | 7    | 8 | 0.500   | 0.250  | 0.033 | 0.250 |
| visible-MPC   | combined      | 0.20   | 0    | 8 | 0.875   | 0.125  | 0.044 | 0.000 |
| visible-MPC   | combined      | 0.20   | 1    | 8 | 0.625   | 0.125  | 0.040 | 0.000 |
| visible-MPC   | combined      | 0.20   | 2    | 8 | 0.500   | 0.375  | 0.042 | 0.000 |
| visible-MPC   | combined      | 0.20   | 3    | 8 | 0.625   | 0.125  | 0.037 | 0.000 |
| visible-MPC   | combined      | 0.20   | 4    | 8 | 0.875   | 0.125  | 0.035 | 0.000 |
| visible-MPC   | combined      | 0.20   | 5    | 8 | 0.625   | 0.000  | 0.040 | 0.000 |
| visible-MPC   | combined      | 0.20   | 6    | 8 | 0.375   | 0.250  | 0.035 | 0.000 |
| visible-MPC   | combined      | 0.20   | 7    | 8 | 0.500   | 0.250  | 0.035 | 0.000 |
| ACM-v4        | hidden-strain | 0.10   | 0    | 8 | 0.500   | 0.125  | 0.008 | 0.500 |
| ACM-v4        | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.006 | 0.625 |
| ACM-v4        | hidden-strain | 0.10   | 2    | 8 | 0.375   | 0.250  | 0.008 | 0.125 |
| ACM-v4        | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.008 | 0.375 |
| ACM-v4        | hidden-strain | 0.10   | 4    | 8 | 0.375   | 0.250  | 0.006 | 0.625 |
| ACM-v4        | hidden-strain | 0.10   | 5    | 8 | 0.375   | 0.375  | 0.008 | 0.500 |

Table 11 continued

| method         | split         | budget | seed | n | success | damage | clip  | diag  |
|----------------|---------------|--------|------|---|---------|--------|-------|-------|
| ACM-v4         | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.125  | 0.004 | 0.750 |
| ACM-v4         | hidden-strain | 0.10   | 7    | 8 | 0.625   | 0.125  | 0.006 | 0.375 |
| ACM-v5         | hidden-strain | 0.10   | 0    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| ACM-v5         | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.006 | 0.000 |
| ACM-v5         | hidden-strain | 0.10   | 2    | 8 | 0.500   | 0.125  | 0.010 | 0.000 |
| ACM-v5         | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.125  | 0.006 | 0.000 |
| ACM-v5         | hidden-strain | 0.10   | 4    | 8 | 0.375   | 0.250  | 0.013 | 0.000 |
| ACM-v5         | hidden-strain | 0.10   | 5    | 8 | 0.375   | 0.375  | 0.013 | 0.000 |
| ACM-v5         | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.125  | 0.010 | 0.000 |
| ACM-v5         | hidden-strain | 0.10   | 7    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 0    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.010 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 4    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 5    | 8 | 0.500   | 0.375  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.10   | 7    | 8 | 0.625   | 0.250  | 0.013 | 0.000 |
| ensemble       | hidden-strain | 0.10   | 0    | 8 | 0.500   | 0.125  | 0.008 | 0.500 |
| ensemble       | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.006 | 0.625 |
| ensemble       | hidden-strain | 0.10   | 2    | 8 | 0.250   | 0.375  | 0.004 | 0.125 |
| ensemble       | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.125  | 0.004 | 0.375 |
| ensemble       | hidden-strain | 0.10   | 4    | 8 | 0.375   | 0.250  | 0.006 | 0.500 |
| ensemble       | hidden-strain | 0.10   | 5    | 8 | 0.375   | 0.375  | 0.006 | 0.500 |
| ensemble       | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.000  | 0.002 | 0.750 |
| ensemble       | hidden-strain | 0.10   | 7    | 8 | 0.625   | 0.125  | 0.006 | 0.500 |
| oracle         | hidden-strain | 0.10   | 0    | 8 | 0.750   | 0.000  | 0.002 | 0.000 |
| oracle         | hidden-strain | 0.10   | 1    | 8 | 1.000   | 0.000  | 0.002 | 0.000 |
| oracle         | hidden-strain | 0.10   | 2    | 8 | 1.000   | 0.000  | 0.004 | 0.000 |
| oracle         | hidden-strain | 0.10   | 3    | 8 | 1.000   | 0.000  | 0.000 | 0.000 |

Table 11 continued



| method        | split         | budget | seed | n | success | damage | clip  | diag  |
|---------------|---------------|--------|------|---|---------|--------|-------|-------|
| oracle        | hidden-strain | 0.10   | 4    | 8 | 1.000   | 0.000  | 0.000 | 0.000 |
| oracle        | hidden-strain | 0.10   | 5    | 8 | 0.625   | 0.125  | 0.004 | 0.000 |
| oracle        | hidden-strain | 0.10   | 6    | 8 | 1.000   | 0.000  | 0.004 | 0.000 |
| oracle        | hidden-strain | 0.10   | 7    | 8 | 0.875   | 0.000  | 0.000 | 0.000 |
| particle      | hidden-strain | 0.10   | 0    | 8 | 0.500   | 0.250  | 0.010 | 0.125 |
| particle      | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.250  | 0.010 | 0.250 |
| particle      | hidden-strain | 0.10   | 2    | 8 | 0.375   | 0.250  | 0.010 | 0.000 |
| particle      | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.010 | 0.375 |
| particle      | hidden-strain | 0.10   | 4    | 8 | 0.375   | 0.250  | 0.008 | 0.375 |
| particle      | hidden-strain | 0.10   | 5    | 8 | 0.375   | 0.375  | 0.013 | 0.250 |
| particle      | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.250  | 0.010 | 0.500 |
| particle      | hidden-strain | 0.10   | 7    | 8 | 0.625   | 0.250  | 0.010 | 0.375 |
| robust-CEM    | hidden-strain | 0.10   | 0    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| robust-CEM    | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.125  | 0.006 | 0.125 |
| robust-CEM    | hidden-strain | 0.10   | 2    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| robust-CEM    | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.125  | 0.008 | 0.125 |
| robust-CEM    | hidden-strain | 0.10   | 4    | 8 | 0.375   | 0.250  | 0.010 | 0.125 |
| robust-CEM    | hidden-strain | 0.10   | 5    | 8 | 0.375   | 0.375  | 0.013 | 0.000 |
| robust-CEM    | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.125  | 0.010 | 0.250 |
| robust-CEM    | hidden-strain | 0.10   | 7    | 8 | 0.750   | 0.250  | 0.010 | 0.125 |
| safe-particle | hidden-strain | 0.10   | 0    | 8 | 0.500   | 0.250  | 0.010 | 0.000 |
| safe-particle | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| safe-particle | hidden-strain | 0.10   | 2    | 8 | 0.500   | 0.125  | 0.010 | 0.000 |
| safe-particle | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| safe-particle | hidden-strain | 0.10   | 4    | 8 | 0.375   | 0.250  | 0.013 | 0.000 |
| safe-particle | hidden-strain | 0.10   | 5    | 8 | 0.375   | 0.375  | 0.013 | 0.000 |
| safe-particle | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| safe-particle | hidden-strain | 0.10   | 7    | 8 | 0.625   | 0.250  | 0.013 | 0.000 |
| visible-MPC   | hidden-strain | 0.10   | 0    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| visible-MPC   | hidden-strain | 0.10   | 1    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |

Table 11 continued

| method         | split         | budget | seed | n | success | damage | clip  | diag  |
|----------------|---------------|--------|------|---|---------|--------|-------|-------|
| visible-MPC    | hidden-strain | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| visible-MPC    | hidden-strain | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.010 | 0.000 |
| visible-MPC    | hidden-strain | 0.10   | 4    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| visible-MPC    | hidden-strain | 0.10   | 5    | 8 | 0.500   | 0.375  | 0.013 | 0.000 |
| visible-MPC    | hidden-strain | 0.10   | 6    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| visible-MPC    | hidden-strain | 0.10   | 7    | 8 | 0.625   | 0.250  | 0.013 | 0.000 |
| ACM-v4         | hidden-strain | 0.20   | 0    | 8 | 0.500   | 0.375  | 0.010 | 0.875 |
| ACM-v4         | hidden-strain | 0.20   | 1    | 8 | 0.625   | 0.250  | 0.008 | 0.875 |
| ACM-v4         | hidden-strain | 0.20   | 2    | 8 | 0.125   | 0.750  | 0.010 | 1.000 |
| ACM-v4         | hidden-strain | 0.20   | 3    | 8 | 0.500   | 0.375  | 0.008 | 0.875 |
| ACM-v4         | hidden-strain | 0.20   | 4    | 8 | 0.500   | 0.375  | 0.010 | 1.000 |
| ACM-v4         | hidden-strain | 0.20   | 5    | 8 | 0.375   | 0.625  | 0.008 | 1.000 |
| ACM-v4         | hidden-strain | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.006 | 1.000 |
| ACM-v4         | hidden-strain | 0.20   | 7    | 8 | 0.625   | 0.375  | 0.013 | 1.000 |
| ACM-v5         | hidden-strain | 0.20   | 0    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| ACM-v5         | hidden-strain | 0.20   | 1    | 8 | 0.625   | 0.125  | 0.006 | 0.000 |
| ACM-v5         | hidden-strain | 0.20   | 2    | 8 | 0.500   | 0.125  | 0.010 | 0.000 |
| ACM-v5         | hidden-strain | 0.20   | 3    | 8 | 0.500   | 0.125  | 0.006 | 0.000 |
| ACM-v5         | hidden-strain | 0.20   | 4    | 8 | 0.375   | 0.250  | 0.013 | 0.000 |
| ACM-v5         | hidden-strain | 0.20   | 5    | 8 | 0.375   | 0.375  | 0.013 | 0.000 |
| ACM-v5         | hidden-strain | 0.20   | 6    | 8 | 0.500   | 0.125  | 0.010 | 0.125 |
| ACM-v5         | hidden-strain | 0.20   | 7    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 0    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 1    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 2    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 3    | 8 | 0.500   | 0.250  | 0.010 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 4    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 5    | 8 | 0.500   | 0.375  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| damage-visible | hidden-strain | 0.20   | 7    | 8 | 0.625   | 0.250  | 0.013 | 0.000 |

Table 11 continued

| method     | split         | budget | seed | n | success | damage | clip  | diag  |
|------------|---------------|--------|------|---|---------|--------|-------|-------|
| ensemble   | hidden-strain | 0.20   | 0    | 8 | 0.500   | 0.375  | 0.008 | 1.000 |
| ensemble   | hidden-strain | 0.20   | 1    | 8 | 0.625   | 0.250  | 0.008 | 0.875 |
| ensemble   | hidden-strain | 0.20   | 2    | 8 | 0.125   | 0.750  | 0.010 | 1.000 |
| ensemble   | hidden-strain | 0.20   | 3    | 8 | 0.500   | 0.375  | 0.008 | 0.875 |
| ensemble   | hidden-strain | 0.20   | 4    | 8 | 0.500   | 0.375  | 0.010 | 1.000 |
| ensemble   | hidden-strain | 0.20   | 5    | 8 | 0.375   | 0.625  | 0.008 | 1.000 |
| ensemble   | hidden-strain | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.006 | 1.000 |
| ensemble   | hidden-strain | 0.20   | 7    | 8 | 0.625   | 0.375  | 0.013 | 1.000 |
| oracle     | hidden-strain | 0.20   | 0    | 8 | 0.750   | 0.000  | 0.002 | 0.000 |
| oracle     | hidden-strain | 0.20   | 1    | 8 | 1.000   | 0.000  | 0.002 | 0.000 |
| oracle     | hidden-strain | 0.20   | 2    | 8 | 1.000   | 0.000  | 0.004 | 0.000 |
| oracle     | hidden-strain | 0.20   | 3    | 8 | 1.000   | 0.000  | 0.000 | 0.000 |
| oracle     | hidden-strain | 0.20   | 4    | 8 | 1.000   | 0.000  | 0.000 | 0.000 |
| oracle     | hidden-strain | 0.20   | 5    | 8 | 0.625   | 0.125  | 0.004 | 0.000 |
| oracle     | hidden-strain | 0.20   | 6    | 8 | 1.000   | 0.000  | 0.004 | 0.000 |
| oracle     | hidden-strain | 0.20   | 7    | 8 | 0.875   | 0.000  | 0.000 | 0.000 |
| particle   | hidden-strain | 0.20   | 0    | 8 | 0.500   | 0.375  | 0.010 | 0.875 |
| particle   | hidden-strain | 0.20   | 1    | 8 | 0.500   | 0.375  | 0.010 | 1.000 |
| particle   | hidden-strain | 0.20   | 2    | 8 | 0.125   | 0.750  | 0.010 | 0.875 |
| particle   | hidden-strain | 0.20   | 3    | 8 | 0.500   | 0.375  | 0.008 | 0.875 |
| particle   | hidden-strain | 0.20   | 4    | 8 | 0.500   | 0.375  | 0.010 | 0.875 |
| particle   | hidden-strain | 0.20   | 5    | 8 | 0.375   | 0.625  | 0.008 | 1.000 |
| particle   | hidden-strain | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.006 | 1.000 |
| particle   | hidden-strain | 0.20   | 7    | 8 | 0.625   | 0.375  | 0.013 | 1.000 |
| robust-CEM | hidden-strain | 0.20   | 0    | 8 | 0.500   | 0.250  | 0.010 | 0.375 |
| robust-CEM | hidden-strain | 0.20   | 1    | 8 | 0.625   | 0.250  | 0.008 | 0.500 |
| robust-CEM | hidden-strain | 0.20   | 2    | 8 | 0.250   | 0.625  | 0.013 | 0.500 |
| robust-CEM | hidden-strain | 0.20   | 3    | 8 | 0.500   | 0.375  | 0.008 | 0.750 |
| robust-CEM | hidden-strain | 0.20   | 4    | 8 | 0.500   | 0.250  | 0.008 | 0.500 |
| robust-CEM | hidden-strain | 0.20   | 5    | 8 | 0.250   | 0.625  | 0.008 | 0.500 |

Table 11 continued

| method        | split            | budget | seed | n | success | damage | clip  | diag  |
|---------------|------------------|--------|------|---|---------|--------|-------|-------|
| robust-CEM    | hidden-strain    | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.013 | 0.625 |
| robust-CEM    | hidden-strain    | 0.20   | 7    | 8 | 0.750   | 0.250  | 0.010 | 0.625 |
| safe-particle | hidden-strain    | 0.20   | 0    | 8 | 0.500   | 0.250  | 0.010 | 0.000 |
| safe-particle | hidden-strain    | 0.20   | 1    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| safe-particle | hidden-strain    | 0.20   | 2    | 8 | 0.500   | 0.125  | 0.010 | 0.000 |
| safe-particle | hidden-strain    | 0.20   | 3    | 8 | 0.500   | 0.125  | 0.008 | 0.000 |
| safe-particle | hidden-strain    | 0.20   | 4    | 8 | 0.375   | 0.250  | 0.013 | 0.000 |
| safe-particle | hidden-strain    | 0.20   | 5    | 8 | 0.375   | 0.375  | 0.013 | 0.000 |
| safe-particle | hidden-strain    | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.013 | 0.125 |
| safe-particle | hidden-strain    | 0.20   | 7    | 8 | 0.625   | 0.250  | 0.013 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 0    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 1    | 8 | 0.625   | 0.250  | 0.010 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 2    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 3    | 8 | 0.500   | 0.250  | 0.010 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 4    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 5    | 8 | 0.500   | 0.375  | 0.013 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.013 | 0.000 |
| visible-MPC   | hidden-strain    | 0.20   | 7    | 8 | 0.625   | 0.250  | 0.013 | 0.000 |
| ACM-v4        | occluded-contact | 0.10   | 0    | 8 | 0.375   | 0.250  | 0.015 | 0.000 |
| ACM-v4        | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.250  | 0.013 | 0.000 |
| ACM-v4        | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.013 | 0.125 |
| ACM-v4        | occluded-contact | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| ACM-v4        | occluded-contact | 0.10   | 4    | 8 | 0.625   | 0.000  | 0.013 | 0.125 |
| ACM-v4        | occluded-contact | 0.10   | 5    | 8 | 0.375   | 0.250  | 0.010 | 0.125 |
| ACM-v4        | occluded-contact | 0.10   | 6    | 8 | 0.375   | 0.375  | 0.013 | 0.250 |
| ACM-v4        | occluded-contact | 0.10   | 7    | 8 | 0.000   | 0.500  | 0.015 | 0.000 |
| ACM-v5        | occluded-contact | 0.10   | 0    | 8 | 0.375   | 0.250  | 0.013 | 0.000 |
| ACM-v5        | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| ACM-v5        | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| ACM-v5        | occluded-contact | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |

Table 11 continued

| method         | split            | budget | seed | n | success | damage | clip  | diag  |
|----------------|------------------|--------|------|---|---------|--------|-------|-------|
| ACM-v5         | occluded-contact | 0.10   | 4    | 8 | 0.625   | 0.000  | 0.013 | 0.000 |
| ACM-v5         | occluded-contact | 0.10   | 5    | 8 | 0.375   | 0.250  | 0.017 | 0.000 |
| ACM-v5         | occluded-contact | 0.10   | 6    | 8 | 0.250   | 0.500  | 0.015 | 0.000 |
| ACM-v5         | occluded-contact | 0.10   | 7    | 8 | 0.125   | 0.375  | 0.010 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 0    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 3    | 8 | 0.625   | 0.250  | 0.017 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 4    | 8 | 0.750   | 0.250  | 0.021 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 5    | 8 | 0.500   | 0.250  | 0.017 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 6    | 8 | 0.500   | 0.250  | 0.021 | 0.000 |
| damage-visible | occluded-contact | 0.10   | 7    | 8 | 0.250   | 0.500  | 0.017 | 0.000 |
| ensemble       | occluded-contact | 0.10   | 0    | 8 | 0.250   | 0.375  | 0.013 | 0.000 |
| ensemble       | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.125  | 0.008 | 0.000 |
| ensemble       | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.125  | 0.010 | 0.125 |
| ensemble       | occluded-contact | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.013 | 0.125 |
| ensemble       | occluded-contact | 0.10   | 4    | 8 | 0.625   | 0.000  | 0.013 | 0.125 |
| ensemble       | occluded-contact | 0.10   | 5    | 8 | 0.375   | 0.250  | 0.010 | 0.125 |
| ensemble       | occluded-contact | 0.10   | 6    | 8 | 0.375   | 0.375  | 0.013 | 0.250 |
| ensemble       | occluded-contact | 0.10   | 7    | 8 | 0.000   | 0.375  | 0.010 | 0.000 |
| oracle         | occluded-contact | 0.10   | 0    | 8 | 0.875   | 0.000  | 0.008 | 0.000 |
| oracle         | occluded-contact | 0.10   | 1    | 8 | 0.750   | 0.000  | 0.006 | 0.000 |
| oracle         | occluded-contact | 0.10   | 2    | 8 | 1.000   | 0.000  | 0.006 | 0.000 |
| oracle         | occluded-contact | 0.10   | 3    | 8 | 0.875   | 0.000  | 0.004 | 0.000 |
| oracle         | occluded-contact | 0.10   | 4    | 8 | 1.000   | 0.000  | 0.006 | 0.000 |
| oracle         | occluded-contact | 0.10   | 5    | 8 | 0.750   | 0.000  | 0.004 | 0.000 |
| oracle         | occluded-contact | 0.10   | 6    | 8 | 1.000   | 0.000  | 0.008 | 0.000 |
| oracle         | occluded-contact | 0.10   | 7    | 8 | 0.750   | 0.000  | 0.002 | 0.000 |
| particle       | occluded-contact | 0.10   | 0    | 8 | 0.375   | 0.250  | 0.015 | 0.000 |
| particle       | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.500  | 0.017 | 0.125 |

Table 11 continued

| method        | split            | budget | seed | n | success | damage | clip  | diag  |
|---------------|------------------|--------|------|---|---------|--------|-------|-------|
| particle      | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| particle      | occluded-contact | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.017 | 0.000 |
| particle      | occluded-contact | 0.10   | 4    | 8 | 0.625   | 0.125  | 0.017 | 0.000 |
| particle      | occluded-contact | 0.10   | 5    | 8 | 0.375   | 0.250  | 0.017 | 0.000 |
| particle      | occluded-contact | 0.10   | 6    | 8 | 0.375   | 0.375  | 0.019 | 0.000 |
| particle      | occluded-contact | 0.10   | 7    | 8 | 0.125   | 0.625  | 0.017 | 0.250 |
| robust-CEM    | occluded-contact | 0.10   | 0    | 8 | 0.375   | 0.250  | 0.015 | 0.000 |
| robust-CEM    | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| robust-CEM    | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| robust-CEM    | occluded-contact | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| robust-CEM    | occluded-contact | 0.10   | 4    | 8 | 0.625   | 0.000  | 0.013 | 0.000 |
| robust-CEM    | occluded-contact | 0.10   | 5    | 8 | 0.375   | 0.250  | 0.017 | 0.000 |
| robust-CEM    | occluded-contact | 0.10   | 6    | 8 | 0.375   | 0.375  | 0.019 | 0.000 |
| robust-CEM    | occluded-contact | 0.10   | 7    | 8 | 0.125   | 0.500  | 0.017 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 0    | 8 | 0.375   | 0.250  | 0.015 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 3    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 4    | 8 | 0.625   | 0.125  | 0.017 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 5    | 8 | 0.375   | 0.250  | 0.017 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 6    | 8 | 0.375   | 0.375  | 0.019 | 0.000 |
| safe-particle | occluded-contact | 0.10   | 7    | 8 | 0.125   | 0.500  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 0    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 3    | 8 | 0.625   | 0.250  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 4    | 8 | 0.750   | 0.250  | 0.021 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 5    | 8 | 0.500   | 0.250  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 6    | 8 | 0.500   | 0.250  | 0.021 | 0.000 |
| visible-MPC   | occluded-contact | 0.10   | 7    | 8 | 0.250   | 0.500  | 0.017 | 0.000 |

Table 11 continued

| method         | split            | budget | seed | n | success | damage | clip  | diag  |
|----------------|------------------|--------|------|---|---------|--------|-------|-------|
| ACM-v4         | occluded-contact | 0.20   | 0    | 8 | 0.125   | 0.625  | 0.008 | 0.875 |
| ACM-v4         | occluded-contact | 0.20   | 1    | 8 | 0.250   | 0.625  | 0.010 | 1.000 |
| ACM-v4         | occluded-contact | 0.20   | 2    | 8 | 0.250   | 0.500  | 0.010 | 1.000 |
| ACM-v4         | occluded-contact | 0.20   | 3    | 8 | 0.500   | 0.375  | 0.013 | 0.875 |
| ACM-v4         | occluded-contact | 0.20   | 4    | 8 | 0.625   | 0.375  | 0.010 | 1.000 |
| ACM-v4         | occluded-contact | 0.20   | 5    | 8 | 0.375   | 0.500  | 0.010 | 0.875 |
| ACM-v4         | occluded-contact | 0.20   | 6    | 8 | 0.250   | 0.625  | 0.013 | 1.000 |
| ACM-v4         | occluded-contact | 0.20   | 7    | 8 | 0.375   | 0.500  | 0.010 | 0.750 |
| ACM-v5         | occluded-contact | 0.20   | 0    | 8 | 0.375   | 0.250  | 0.013 | 0.000 |
| ACM-v5         | occluded-contact | 0.20   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| ACM-v5         | occluded-contact | 0.20   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| ACM-v5         | occluded-contact | 0.20   | 3    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| ACM-v5         | occluded-contact | 0.20   | 4    | 8 | 0.625   | 0.000  | 0.013 | 0.125 |
| ACM-v5         | occluded-contact | 0.20   | 5    | 8 | 0.375   | 0.250  | 0.017 | 0.000 |
| ACM-v5         | occluded-contact | 0.20   | 6    | 8 | 0.250   | 0.500  | 0.015 | 0.000 |
| ACM-v5         | occluded-contact | 0.20   | 7    | 8 | 0.125   | 0.375  | 0.010 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 0    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 3    | 8 | 0.625   | 0.250  | 0.017 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 4    | 8 | 0.750   | 0.250  | 0.021 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 5    | 8 | 0.500   | 0.250  | 0.017 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.021 | 0.000 |
| damage-visible | occluded-contact | 0.20   | 7    | 8 | 0.250   | 0.500  | 0.017 | 0.000 |
| ensemble       | occluded-contact | 0.20   | 0    | 8 | 0.125   | 0.625  | 0.006 | 0.875 |
| ensemble       | occluded-contact | 0.20   | 1    | 8 | 0.250   | 0.625  | 0.010 | 1.000 |
| ensemble       | occluded-contact | 0.20   | 2    | 8 | 0.250   | 0.500  | 0.010 | 0.875 |
| ensemble       | occluded-contact | 0.20   | 3    | 8 | 0.500   | 0.375  | 0.013 | 1.000 |
| ensemble       | occluded-contact | 0.20   | 4    | 8 | 0.625   | 0.375  | 0.010 | 1.000 |
| ensemble       | occluded-contact | 0.20   | 5    | 8 | 0.375   | 0.500  | 0.010 | 0.875 |

Table 11 continued

| method        | split            | budget | seed | n | success | damage | clip  | diag  |
|---------------|------------------|--------|------|---|---------|--------|-------|-------|
| ensemble      | occluded-contact | 0.20   | 6    | 8 | 0.250   | 0.625  | 0.013 | 1.000 |
| ensemble      | occluded-contact | 0.20   | 7    | 8 | 0.375   | 0.500  | 0.010 | 0.750 |
| oracle        | occluded-contact | 0.20   | 0    | 8 | 0.875   | 0.000  | 0.008 | 0.000 |
| oracle        | occluded-contact | 0.20   | 1    | 8 | 0.750   | 0.000  | 0.006 | 0.000 |
| oracle        | occluded-contact | 0.20   | 2    | 8 | 1.000   | 0.000  | 0.006 | 0.000 |
| oracle        | occluded-contact | 0.20   | 3    | 8 | 0.875   | 0.000  | 0.004 | 0.000 |
| oracle        | occluded-contact | 0.20   | 4    | 8 | 1.000   | 0.000  | 0.006 | 0.000 |
| oracle        | occluded-contact | 0.20   | 5    | 8 | 0.750   | 0.000  | 0.004 | 0.000 |
| oracle        | occluded-contact | 0.20   | 6    | 8 | 1.000   | 0.000  | 0.008 | 0.000 |
| oracle        | occluded-contact | 0.20   | 7    | 8 | 0.750   | 0.000  | 0.002 | 0.000 |
| particle      | occluded-contact | 0.20   | 0    | 8 | 0.125   | 0.625  | 0.006 | 1.000 |
| particle      | occluded-contact | 0.20   | 1    | 8 | 0.250   | 0.625  | 0.010 | 1.000 |
| particle      | occluded-contact | 0.20   | 2    | 8 | 0.250   | 0.500  | 0.010 | 1.000 |
| particle      | occluded-contact | 0.20   | 3    | 8 | 0.500   | 0.375  | 0.013 | 1.000 |
| particle      | occluded-contact | 0.20   | 4    | 8 | 0.625   | 0.375  | 0.010 | 1.000 |
| particle      | occluded-contact | 0.20   | 5    | 8 | 0.375   | 0.500  | 0.008 | 1.000 |
| particle      | occluded-contact | 0.20   | 6    | 8 | 0.250   | 0.625  | 0.013 | 1.000 |
| particle      | occluded-contact | 0.20   | 7    | 8 | 0.500   | 0.500  | 0.008 | 1.000 |
| robust-CEM    | occluded-contact | 0.20   | 0    | 8 | 0.125   | 0.500  | 0.010 | 0.500 |
| robust-CEM    | occluded-contact | 0.20   | 1    | 8 | 0.375   | 0.500  | 0.017 | 0.375 |
| robust-CEM    | occluded-contact | 0.20   | 2    | 8 | 0.375   | 0.375  | 0.010 | 0.625 |
| robust-CEM    | occluded-contact | 0.20   | 3    | 8 | 0.500   | 0.250  | 0.013 | 0.500 |
| robust-CEM    | occluded-contact | 0.20   | 4    | 8 | 0.625   | 0.250  | 0.017 | 0.375 |
| robust-CEM    | occluded-contact | 0.20   | 5    | 8 | 0.250   | 0.500  | 0.010 | 0.750 |
| robust-CEM    | occluded-contact | 0.20   | 6    | 8 | 0.500   | 0.375  | 0.015 | 0.500 |
| robust-CEM    | occluded-contact | 0.20   | 7    | 8 | 0.375   | 0.375  | 0.013 | 0.375 |
| safe-particle | occluded-contact | 0.20   | 0    | 8 | 0.375   | 0.250  | 0.015 | 0.000 |
| safe-particle | occluded-contact | 0.20   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| safe-particle | occluded-contact | 0.20   | 2    | 8 | 0.500   | 0.250  | 0.013 | 0.125 |
| safe-particle | occluded-contact | 0.20   | 3    | 8 | 0.500   | 0.250  | 0.013 | 0.250 |

Table 11 continued

| method        | split            | budget | seed | n | success | damage | clip  | diag  |
|---------------|------------------|--------|------|---|---------|--------|-------|-------|
| safe-particle | occluded-contact | 0.20   | 4    | 8 | 0.625   | 0.125  | 0.017 | 0.125 |
| safe-particle | occluded-contact | 0.20   | 5    | 8 | 0.375   | 0.250  | 0.015 | 0.125 |
| safe-particle | occluded-contact | 0.20   | 6    | 8 | 0.375   | 0.375  | 0.015 | 0.250 |
| safe-particle | occluded-contact | 0.20   | 7    | 8 | 0.125   | 0.500  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 0    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 1    | 8 | 0.375   | 0.375  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 2    | 8 | 0.500   | 0.250  | 0.015 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 3    | 8 | 0.625   | 0.250  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 4    | 8 | 0.750   | 0.250  | 0.021 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 5    | 8 | 0.500   | 0.250  | 0.017 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 6    | 8 | 0.500   | 0.250  | 0.021 | 0.000 |
| visible-MPC   | occluded-contact | 0.20   | 7    | 8 | 0.250   | 0.500  | 0.017 | 0.000 |

## F REPRODUCIBILITY CHECKLIST

Table 12: Artifact and reproducibility checklist.

| item                                    | status                                                                                              |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| CPU-only execution                      | all experiments use NumPy, matplotlib, scikit-learn Ridge models, and a local mass-spring simulator |
| RAM-light execution                     | single-process run; no checkpoints or large neural tensors                                          |
| No Desktop PDF                          | final numbered PDF is copied only to Downloads                                                      |
| Bright citation boxes                   | hyperref uses green citation borders, orange internal links, and blue URL borders                   |
| Frozen main split                       | combined_memory_stress is the primary gate                                                          |
| Frozen aggregate split                  | hard_regime averages hidden_strain, occluded_contact, material_shift, and combined stress           |
| Frozen fixed-risk budgets               | 0.10, 0.20, 0.30, and 0.40                                                                          |
| Oracle disclosed                        | oracle_latent_state is an upper bound, not a deployable method                                      |
| Hardware limitation disclosed           | no real-robot experiments are claimed                                                               |
| External benchmark limitation disclosed | no public deformable benchmark validation is claimed                                                |
| Negative cases retained                 | scenario-level failures remain in the artifact                                                      |
| Ablation gate retained                  | full v5 must beat its own removals                                                                  |
| Stress gate retained                    | maximum stress is reported even when unfavorable                                                    |
| Paired comparisons retained             | success differences are computed over matched seeds                                                 |
| Safety reported                         | damage, force clipping, contact, and diagnostics are not hidden                                     |
| No fake citations                       | bibliography entries are real sources checked against primary pages                                 |

## G PROTOCOL NOTES

The simulator is intentionally small enough to rerun on CPU. Each rollout records seed, scenario, split, object type, chosen action, true and estimated memory, uncertainty, visible fraction, success, shape error, memory error, damage, stretch, contact, force clipping, strain energy, progress, diagnostic count, actual failures, predicted failures, and a compact trajectory trace. The raw rollout file is not typeset in full because it is thousands of rows, but the seed-level aggregates, fixed-risk aggregates, stress aggregates, ablations, pairwise statistics, and negative cases are included above. The code path that produced the paper is:

```
python src\run_experiment.py
python scripts\generate_manuscript.py
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

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